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Week 1 | 6.01~6.07

Participants: Nianyi Cheng, Xiaofan Luo, Jiayu Wang, Atita Sartsasi, Qingxin Ma, Flora Da, Chenming Jin, Siqi Ding, Ziyu Zhong, Maoyuan Sun, Peilin Zhu

Project Background and Motivation

Poly(ethylene terephthalate) (PET) is one of the most widely used synthetic polyesters and is extensively employed in packaging, textiles, bottles, and consumer products. The same properties that make PET attractive in use—including mechanical strength, chemical resistance, and hydrolytic stability—also contribute to its persistence after disposal. Low recycling rates and limited end-of-life pathways have therefore made PET an important target for the development of alternative recycling technologies.

Enzymatic PET depolymerization offers a complementary strategy to conventional mechanical and thermochemical recycling. Rather than simply reforming used plastic into materials of progressively lower quality, enzymatic hydrolysis can cleave PET ester bonds and recover soluble chemical building blocks that can potentially re-enter polymer-production cycles. However, practical PET recycling is not determined by enzyme activity alone. Polymer morphology, substrate accessibility, enzyme stability, reaction temperature, and the molecular organization of the catalytic complex all contribute to the final depolymerization performance.

Our project therefore approached PET-hydrolase engineering at multiple levels. We first considered the physical properties of the PET substrate and the structural architecture of PET hydrolases. We then combined mutation design, molecular simulation, quantum-chemical analysis, and experimental validation to ask not only whether a mutation improves PET hydrolysis, but which molecular bottleneck it relieves.

PET Plastic Pollution and Enzymatic Recycling

PET can be recycled through several fundamentally different routes. Mechanical recycling generally involves sorting, washing, grinding, melting, and re-extrusion. Although widely implemented, repeated mechanical processing can result in deterioration of material properties. Chemical recycling instead depolymerizes PET into smaller chemical species that can be reused as feedstocks. Enzymatic hydrolysis represents a biologically catalyzed version of this depolymerization concept and has attracted particular interest because it can operate under comparatively mild aqueous conditions.

PET hydrolases cleave ester bonds within the polyester chain, generating soluble intermediates and products including bis(2-hydroxyethyl) terephthalate (BHET), mono(2-hydroxyethyl) terephthalate (MHET), terephthalic acid (TPA), and ethylene glycol (EG). Additional BHET- or MHET-hydrolyzing enzymes

can further convert intermediate products toward TPA and EG. Compared with conventional chemical depolymerization processes, enzymatic PET hydrolysis can operate at lower temperatures and atmospheric pressure and provides high product selectivity, although reaction time, enzyme stability, water use, and process economics remain important limitations.

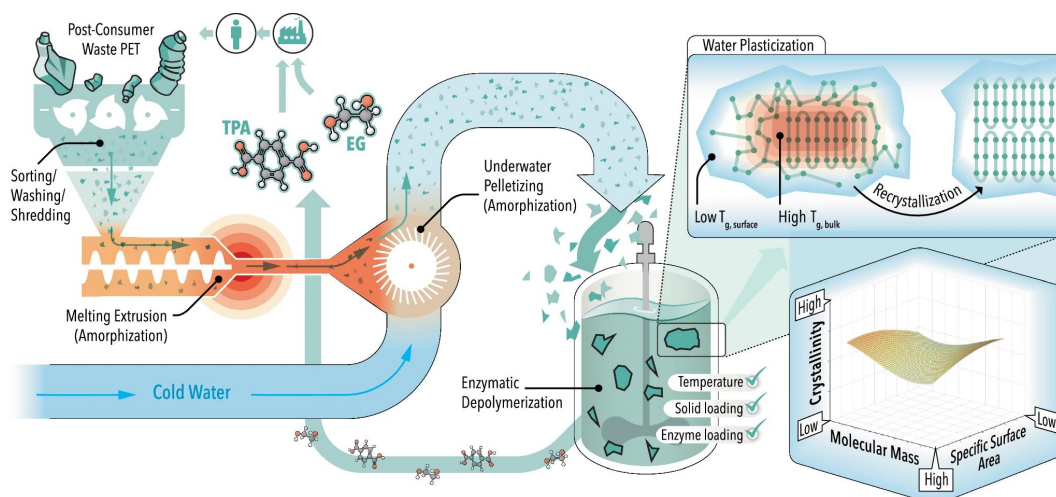


Figure: Schematic representation of PET pretreatment and enzymatic recycling. Amorphization and size reduction increase polymer-chain accessibility and specific surface area, while crystallinity, molecular weight, thermal history, and reaction temperature influence the fraction of PET available for enzymatic attack.

A critical feature of enzymatic PET degradation is that PET is not a conventional soluble small-molecule substrate. It is a semi-crystalline solid polymer, and only polymer chains that are sufficiently accessible to the enzyme can readily enter the catalytic cycle.

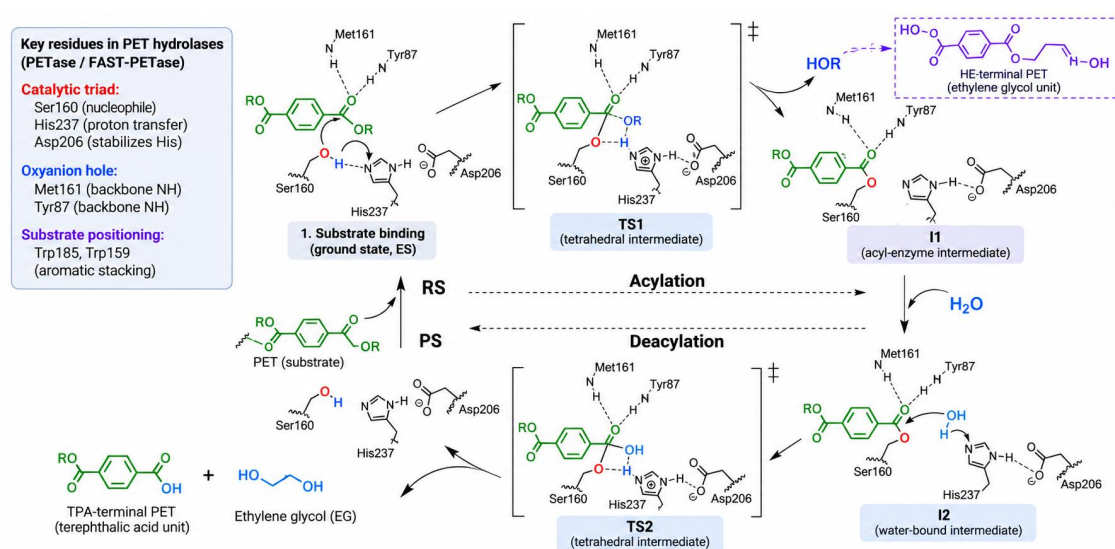
The amorphous and crystalline regions of PET therefore behave very differently. In crystalline regions, polymer chains are tightly packed and highly ordered, reducing access by both the enzyme and water molecules. Amorphous regions contain more disordered and mobile chains and are consequently much more susceptible to enzymatic hydrolysis. Industrial and laboratory pretreatment strategies therefore frequently aim to reduce crystallinity while increasing the specific surface area available to the enzyme.

PET Hydrolases and FAST-PETase

PET hydrolases belong primarily to the α/β -hydrolase family and use a catalytic architecture characteristic of serine hydrolases. In the extensively studied *Ideonella sakaiensis* PETase system, the catalytic triad consists of Ser160, His237, and Asp206. The nucleophilic serine attacks the carbonyl carbon of a PET ester bond, while His237 participates in proton transfer and Asp206 supports the catalytic histidine.

Catalysis also depends on residues outside the catalytic triad. The backbone amides of Met161 and Tyr87 contribute to the oxyanion-hole environment, stabilizing the developing negative charge on the PET carbonyl oxygen. Aromatic residues including Trp185 and Trp159 participate in substrate positioning around the exposed binding cleft. Thus, productive catalysis requires the correct organization of a larger interaction network rather than the catalytic triad alone.

This wider architecture is particularly important for enzyme engineering. Mutations do not need to replace a catalytic residue directly to influence catalysis. Changes in substrate-binding residues, second-shell interactions, hydration, loop dynamics, or local packing can alter how frequently the enzyme reaches a productive configuration.



Schematic Representation of the PETase's Reaction Mechanism

Among engineered PET hydrolases, FAST-PETase provides a particularly useful scaffold for further investigation. FAST-PETase was developed from the IsPETase/ThermoPETase lineage using the structure-based machine-learning method MutCompute. MutCompute identifies residues whose wild-type identities appear suboptimal for their local structural microenvironments and ranks alternative amino acids according to predicted probabilities. Experimental combination of selected substitutions ultimately produced FAST-PETase, containing the additional substitutions S121E, R224Q, and N233K relative to ThermoPETase. The resulting enzyme displayed substantially enhanced PET-hydrolysis activity under the conditions examined in the original engineering study.

Challenges in PET Hydrolase Engineering

PET-hydrolase engineering is a multidimensional optimization problem.

For industrially useful PET depolymerization, high catalytic activity alone is insufficient. Important engineering objectives also include thermostability, expression and solubility, pH tolerance, substrate tolerance, and resistance to product inhibition. An enzyme must remain active over long reaction times, function at high PET loadings, and ideally tolerate a broad range of polymer forms and process conditions.

These requirements can also interact or compete with one another. A mutation that increases structural rigidity may improve thermostability while reducing conformational flexibility required for substrate accommodation. Increasing substrate affinity can improve enzyme–polymer association but may also stabilize a nonproductive binding mode. A globally stable enzyme may still position the reactive PET ester poorly, whereas a locally flexible enzyme may sample catalytically productive states more frequently.

Consequently, macroscopic PET degradation cannot be assigned to a single structural descriptor. The broader mechanistic framework developed for PET hydrolases emphasizes that polymer adsorption, substrate accommodation, chemical reaction, and product release can impose different limitations at different stages.

This has important consequences for computational design.

A favorable docking score does not demonstrate that a substrate is positioned for chemistry.

A low backbone RMSD does not demonstrate that the active site is catalytically optimized.

A large number of enzyme–substrate hydrogen bonds does not necessarily indicate a productive catalytic complex.

Likewise, a single optimized structure cannot represent the complete conformational ensemble sampled by an enzyme–polymer system.

These limitations motivated us to use different computational methods for different physical questions rather than treating any single method as a universal activity predictor.

Week 2 | 6.08~6.15

Participants: Nianyi Cheng, Xiaofan Luo, Jiayu Wang, Atita Sartsasi, Flora Da, Gao Yishan, Chenming Jin, Yueyang Du, Jiahua Wang, Siqi Ding, Peilin Zhu, Maoyuan Sun

From Improved Activity to Mechanistic Understanding

A central motivation of this project was the distinction between observing improved activity and understanding why that activity improves.

PETase catalysis follows the general mechanism of a serine hydrolase. Structural and computational studies support a two-stage process involving acylation and deacylation. During acylation, Ser160 attacks the PET ester carbonyl while proton transfer is mediated by His237 and the developing oxyanion is stabilized by the local oxyanion-hole environment. Detailed transition-path simulations have shown that bond formation, ester cleavage, and proton transfer are strongly coupled rather than being independent events.

Importantly, recent mechanistic simulations of IsPETase predict that deacylation, rather than acylation, is rate-limiting under the modeled conditions.

For this reason, our use of the acylation step should not be interpreted as a claim that acylation universally determines the overall experimental PET-depolymerization rate. Instead, acylation provides a structurally well-defined chemical event through which the consequences of mutation-induced changes in catalytic organization can be compared consistently.

Before acylation can occur, the substrate must first adopt an attack-compatible geometry. The catalytic serine oxygen must approach the ester carbonyl carbon at an appropriate nucleophile–electrophile distance and Bürgi–Dunitz approach angle, while the carbonyl oxygen must be oriented toward the oxyanion hole and the catalytic proton-transfer network must remain appropriately organized. This suggests that catalysis can be considered at two related levels.

The first is geometric accessibility: How frequently does the enzyme–PET ensemble reach a configuration compatible with nucleophics; This can be characterized using geometric pre-reaction-state (PRS) analysis.

The second is chemical productivity: Once a geometrically competent configuration is reached, how well is that state organized for proton transfer, oxyanion stabilization, substrate polarization, and transition-state formation?

A geometric PRS does not automatically guarantee chemical productivity. Configurations satisfying the same Ser–carbonyl distance and attack-angle criteria may still differ in oxyanion-hole engagement, proton-transfer readiness, electrostatic organization, and transition-state stabilization.

Overall Engineering Strategy

The engineering strategy was designed to connect mutation discovery, physical construction, functional validation, and mechanistic interpretation within a single iterative framework.

Rather than organizing the project around individual computational programs or experimental techniques, we organized it around a sequence of biological questions:

Which mutations should we test? Can they be constructed and expressed? Do they improve PET hydrolysis? What physical change explains the improvement? Which bottleneck should be targeted in the next design cycle?

This distinction was important because candidate-generation methods and mechanistic-analysis methods serve different purposes. Machine-learning predictions and docking scores can help identify variants worth testing, but they do not by themselves establish the physical origin of improved catalysis. Likewise, mechanistic simulations are most informative when interpreted together with experimentally validated phenotypes.

Two Complementary Mutation-Design Routes

To reduce dependence on a single design philosophy, candidate generation was performed through two complementary routes.

Route I — Graph-Learning-Guided Design

The first route used structure-aware graph learning to explore mutation possibilities within the three-dimensional FAST-PETase environment.

A GVP-GNN-based strategy was used to identify substitutions compatible with local sequence and structural context. Model-supported candidates were subsequently mapped onto the enzyme structure and filtered according to their relationship with the catalytic pocket and PET-binding region.

This route was particularly useful for identifying mutations that might not be obvious from direct active-site inspection, including second-shell substitutions capable of modifying substrate positioning, local packing, solvation, or conformational dynamics.

The graph-learning route ultimately provided the candidate **Y87Q** for downstream experimental and mechanistic analysis.

Route II — Structure-Based Active-Site Design

The second route began from the opposite direction.

Rather than searching broadly through learned protein sequence–structure relationships, residues surrounding the PET-binding region were explicitly selected and alternative substitutions were systematically enumerated.

Candidate structures were modeled, filtered for structural plausibility, and evaluated by docking with PET-derived substrates. Importantly, docking energy was not treated as a direct prediction of catalytic activity. Candidate poses also had to preserve a physically plausible catalytic orientation of the reacting PET ester.

A reduced set of active-site substitutions was then carried forward into experimental screening, from which S187Y emerged as a principal candidate for mechanistic investigation.

The two routes therefore explored different parts of sequence space:

Graph learning: *Which substitutions are supported by the broader sequence–structure environment?*

Structure-based design: *Which substitutions directly reorganize the local enzyme–substrate environment?* Both routes ultimately converged on the same requirement:

a computationally attractive candidate must still be experimentally validated and mechanistically interpreted.

Computational and Experimental Workflow

The complete project workflow connected progressively more detailed descriptions of the same catalytic problem.

At the structural level, experimentally resolved PET-hydrolase structures were inspected to identify the catalytic triad, substrate-binding region, oxyanion hole, mutation sites, protonation states, and relevant structural features.

PET-derived substrate models were then constructed and subjected to conformational sampling because a flexible PET oligomer cannot be adequately represented by one manually drawn geometry.

Candidate variants from the two design routes were combined with representative PET conformers through molecular docking. Docking provided plausible bound complexes and an initial geometric filter, but was not used as the final catalytic ranking criterion.

Selected systems were then subjected to replicated explicit-solvent molecular dynamics. Global metrics such as RMSD, RMSF, radius of gyration, and secondary structure were first used as structural quality controls. Analysis then moved toward increasingly reaction-specific descriptors, including local solvation, substrate contacts, catalytic distances, and attack geometry.

Structural Preparation of PET Hydrolases

Week 3 | 6.15~6.23

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Selection of Structural Templates

We retrieved experimentally resolved PET-hydrolase structures from the Protein Data Bank and practiced a repeatable inspection workflow rather than relying on visual impressions. For each structure, we checked chain identity, missing residues, alternate conformations, crystallographic ligands, catalytic residues, disulfide bonds, and the local environment surrounding the substrate-binding cleft. Particular attention was paid to PDB 7SH6 for FAST-

PETase and 6THT for the LCC-ICCG scaffold because these structures later served as computational templates.

Structural Inspection with PyMOL and VMD

In PyMOL, we learned to display the overall α/β -hydrolase fold, isolate the catalytic triad, show substrate-contacting residues as sticks, measure distances, and create consistent views for comparison between parent and mutant enzymes. VMD was introduced as the complementary tool for trajectory visualization: a crystal structure provides one coordinate set, whereas VMD allows us to observe how those coordinates change over time during MD. This distinction became central later when we asked whether mutations alter global structure or instead redistribute local conformational states.

In parallel, we learned the fundamentals of VMD (Visual Molecular Dynamics), a molecular visualization and analysis program particularly useful for molecular dynamics simulations.

While PyMOL is highly convenient for inspecting static protein structures and producing clear structural figures, VMD is especially powerful for handling large molecular systems and visualizing simulation trajectories. This made VMD an important tool for the next stages of our computational workflow.

During the training, we practiced loading molecular structures and trajectories, changing molecular representations, selecting specific atoms or residues, and inspecting different components of a simulation system. We also became familiar with displaying proteins, substrates, water molecules, and ions separately, which is useful for checking whether a molecular system has been constructed correctly.

One of the most important advantages of VMD is its ability to visualize structural changes over time. By examining MD trajectories frame by frame, we can observe protein conformational changes, substrate movement, and changes in interactions between molecules throughout a simulation.

VMD also provides a foundation for quantitative trajectory analysis. Measurements such as distances, angles, hydrogen-bond patterns, and structural deviations can be extracted from simulation trajectories to complement visual observations.

PyMOL is a molecular visualization program widely used for the three-dimensional inspection, analysis, and presentation of biological macromolecules, including proteins, nucleic acids, ligands, and molecular complexes. Learning PyMOL provided us with an intuitive way to connect structural information with the possible biological functions of our target proteins.

We first became familiar with loading and manipulating molecular structures. Protein structures in formats such as PDB can be imported directly

into PyMOL and explored interactively by rotating, translating, and zooming the molecular model. We then practiced several commonly used representations, including cartoon, sticks, spheres, and surface.

PyMOL

Molecular Visualization for Structural Analysis and Presentation

1. Overview

PyMOL is a widely used molecular visualization program for the 3D inspection, analysis, and high-quality presentation of biological macromolecules, including proteins, nucleic acids, ligands, and molecular complexes.

Key Features

- ✓ High-resolution 3D visualization
- ✓ Interactive manipulation
- ✓ Flexible representation styles
- ✓ Residue/atom selection and coloring
- ✓ Publication-ready figures

2. Workflow in PyMOL

1. Load structure
Import PDB, MOL2, SDF, etc.

2. Adjust view
Rotate, translate, zoom

3. Choose representation
Cartoon, sticks, surface, etc.

4. Select and color
Highlight specific residues, ligands or regions

5. Analyze and export
Inspect interactions and generate figures

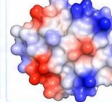
3. Visualization Examples

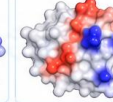
Cartoon representation

Shows overall protein fold and secondary structure.

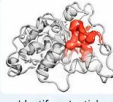
Stick representation

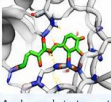
Displays atomic details and key interactions (e.g., catalytic residues).

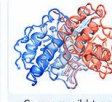
Surface representation

Visualizes molecular surface and binding pockets.

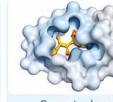
Color by property

Residues can be colored by charge, hydrophobicity, conservation, etc.

4. Applications in Our Project


Identify potential active sites


Analyze substrate-protein interactions


Compare wild-type and mutant structures


Generate clear, publication-ready figures

VMD

Visual Molecular Dynamics for Simulation Visualization and Analysis

1. Overview

VMD (Visual Molecular Dynamics) is a molecular visualization and analysis program designed for large molecular systems, especially molecular dynamics (MD) simulations. It allows users to visualize, analyze, and manipulate biomolecular systems and trajectories efficiently.

Key Features

- ✓ Designed for MD simulations
- ✓ Efficiently handles large systems
- ✓ Visualization of trajectories over time
- ✓ Flexible atom/residue selection
- ✓ Built-in analysis tools (distance, RMSD, H-bonds, etc.)
- ✓ Supports various file formats (PDB, PSF, DCD, XTC, etc.)

2. Workflow in VMD

1. Load system
Import structure and trajectory files (e.g., PDB, PSF, DCD).

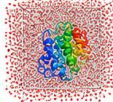
2. Set representation
Display protein, ligand, water, ions separately.

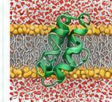
3. Visualize trajectory
Play, pause, or step through frames.

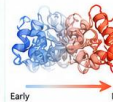
4. Analyze
Measure distances, angles, RMSD, H-bonds, etc.

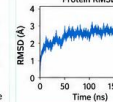
5. Export
Save images, movies, or analysis results.

3. Visualization Examples

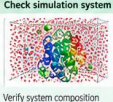
Full simulation system

Protein (cartoon), water (dots), and ions (spheres).

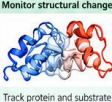
Different representations

Display multiple components with different styles.

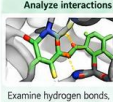
Trajectory visualization

Observe conformational changes over time.

Analysis example: RMSD

Built-in tools for quantitative trajectory analysis.

4. Applications in Our Project

Check simulation system

Verify system composition and structure before simulation.

Monitor structural changes

Track protein and substrate conformational dynamics.

Analyze interactions

Examine hydrogen bonds, distances, and binding behavior during simulation.

Generate movies and figures

Create publication-quality movies and figures.

Different representations emphasize different structural features. Cartoon representations provide a clear overview of protein folds and secondary structures, while stick representations allow closer inspection of individual residues, ligands, and potential intermolecular interactions. Surface representations are particularly useful for examining molecular shape, cavities, and possible substrate-binding regions.

We also learned to select and color specific residues or structural regions using PyMOL commands. For example, catalytic residues, ligands, or residues of particular interest can be highlighted independently from the rest of the protein. This makes it much easier to investigate the spatial relationships between residues that may be distant in the primary sequence but close to one another in the folded protein.

Through this training, we learned that PyMOL and VMD serve complementary roles rather than interchangeable ones. PyMOL is particularly useful for detailed inspection of static structures, residue-level visualization, and preparation of structural figures, whereas VMD is well suited for examining complete simulation systems and analyzing molecular dynamics trajectories.

Together, these tools provide an essential visual and analytical foundation for our subsequent computational work. Before running simulations, they allow us to inspect the quality and composition of our molecular models.

After simulations, they enable us to examine structural changes and interactions over time.

Identification of the Catalytic Center and Substrate-Binding Region

Before performing molecular docking and molecular dynamics (MD) simulations, we first established a reproducible computational workflow for FAST-PETase.

The objectives of this stage were to:

- understand the three-dimensional structure of FAST-PETase and identify structural features relevant to PET hydrolysis;

- assign chemically reasonable protonation states, particularly for residues in and around the catalytic center;

- become familiar with Linux-based high-performance computing (HPC) and remote task submission; standardize directory structures, input files, and analysis scripts to improve computational reproducibility; explore the conformational flexibility of PET oligomers and generate representative PET trimer conformations for subsequent protein–substrate docking.

Together, these preparations established the structural and computational foundation for the later Docking → MD Simulation → Mutation Design → Experimental Validation workflow.

Based on the structural inspection established in Week 1, we selected the FAST-PETase structure as the reference model and proceeded to chemically relevant structure preparation. Particular attention was then given to protonation states around the catalytic center.

Protonation-State Assignment with PROPKA

Before molecular simulation, appropriate protonation states must be assigned to ionizable residues. This is especially important for enzyme simulations because an inappropriate protonation state can alter electrostatic interactions, hydrogen-bond networks, and the geometry of the catalytic center.

We used PROPKA to estimate residue pKa values in the context of the FAST-PETase structure. Protonation states were evaluated under conditions corresponding to approximately pH 8.0, consistent with the conditions selected for our comparative computational analysis.

Special attention was given to titratable residues located within or near the active site.

The catalytic histidine required particularly careful consideration because its protonation state determines which imidazole nitrogen can participate in the proton-transfer network involving the catalytic serine and aspartate. Based on the predicted pKa values and inspection of the local hydrogen-bond

geometry, the catalytic histidine was represented with N δ protonated and N ϵ available for interaction with the catalytic serine in our prepared model.

Importantly, PROPKA predictions were treated as a decision-support tool rather than an automatic assignment. After obtaining the predicted protonation states, residues surrounding the catalytic center were inspected in three dimensions. We checked whether the resulting hydrogen-bond donor/acceptor patterns were chemically reasonable and whether the selected protonation states or histidine tautomers introduced unfavorable local interactions.

The final protonation decisions were documented together with the corresponding prepared structure. This ensured that wild-type FAST-PETase and subsequent mutant systems could be constructed from a clearly defined and reproducible reference state.

Week 4 | 6.24~6.30

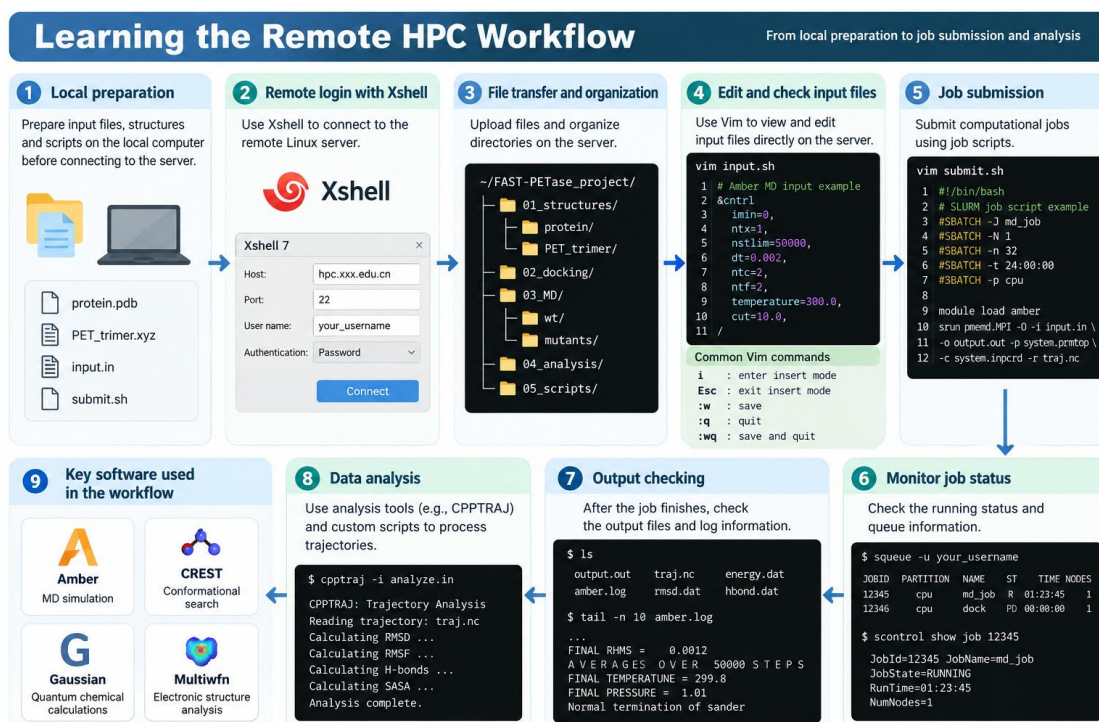
Participants: Nianyi Cheng, Xiaofan Luo, Jiayu Wang, Atita Sartsasi, Jiahua Wang, Qingxin Ma, Gao Yishan, Chenming Jin, Peilin Zhu, Yueyang Du, Haoran Luo, Siqi Ding, Xinglin Zhang, Maoyuan Sun

Reproducible Computational Environment

The computational component of our project required calculations that could not be efficiently performed on a local computer. We therefore established a remote-computing workflow for accessing the high-performance computing platform and submitting computational jobs.

We used Xshell to connect to the remote Linux environment and became familiar with the basic workflow of: local preparation → remote login → file transfer/organization → input inspection → job submission → job monitoring → output checking → trajectory/data analysis

This training was necessary because programs used later in the project, including Amber, CREST, Gaussian, and Multiwfn, were operated primarily through command-line and text-based workflows. Learning to work directly in the Linux environment also improved reproducibility. Instead of relying on a sequence of undocumented graphical operations, simulation parameters and computational procedures could be stored in human-readable input files and scripts.



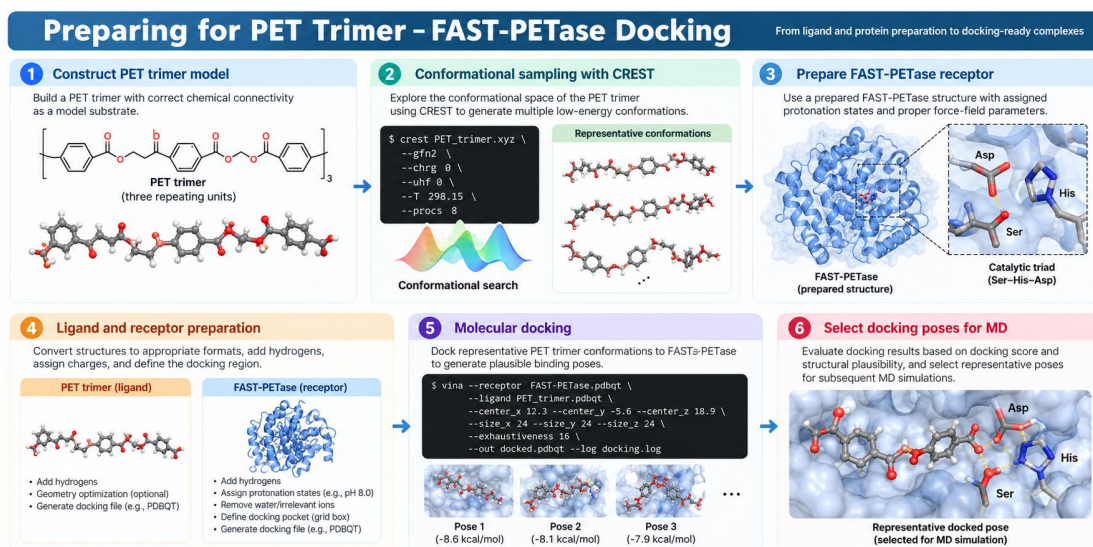
Preparing for PET Trimer–FAST-PETase Docking

The structural preparation of FAST-PETase and conformational sampling of the PET trimer converged in the next stage of our computational workflow: construction of an enzyme–substrate complex. Representative PET trimer conformations generated during conformational sampling were used as candidate ligands for molecular docking with FAST-PETase.

Docking was not treated simply as a search for the pose with the numerically lowest docking score. Because our ultimate goal was to investigate catalytically relevant enzyme–substrate interactions, candidate poses also needed to be evaluated according to their structural plausibility.

In particular, we examined whether the PET chain could be accommodated within the substrate-binding region without severe steric clashes and whether an ester linkage could adopt a reasonable orientation relative to the catalytic center.

Docking therefore served primarily as a method for generating plausible starting enzyme–PET complexes for subsequent molecular dynamics simulations, rather than as definitive evidence for a single substrate-binding conformation.



PET-derived oligomers are considerably more flexible than conventional small-molecule ligands. Therefore, using a single manually constructed ligand geometry could introduce substantial bias into subsequent docking calculations.

To obtain more representative starting structures, we first explored the conformational space of the PET-derived ligand 2HE-MHET2. Conformational sampling generated multiple geometrically distinct structures, allowing us to account for different arrangements of the aromatic rings, ester bonds, and flexible linker regions.

Representative conformations from this ensemble were subsequently used in the docking workflow.

This step was important because our aim was not simply to identify a ligand conformation with a favorable docking score. Instead, we wanted to determine whether PET-derived substrates could adopt orientations compatible with the catalytic machinery of FAST-PETase.

Our computational workflow therefore progressed from: Ligand construction → Conformational sampling → Representative conformers → Molecular docking → Productive-pose filtering rather than relying on a single arbitrary PET-derived structure.

The docking results were first filtered using predefined distance constraints to identify potentially productive binding conformations. In particular, productive poses were defined by the appropriate positioning of one of the four catalytic carbonyl carbons of 2HE-MHET2 relative to the catalytic residue S160, together with additional distance constraints involving residues surrounding the active site. Four representative productive binding orientations (Poses I–IV) were considered to account for the different possible orientations of the ligand within the catalytic pocket. Variants that passed the geometric filters were subsequently ranked according to their predicted

binding free energy. The highest-ranking candidates were then visually inspected to confirm reasonable ligand positioning and active-site interactions. Previously reported unproductive mutations were excluded, and the remaining promising mutants were selected for subsequent in vitro experimental validation.

This workflow demonstrates how structural modelling and molecular docking can be used as a computational pre-screening strategy for PET hydrolase engineering. Rather than selecting mutants based only on docking energy, the study first evaluated whether the PET-derived ligand adopted a catalytically productive geometry in the enzyme active site. Distance-based geometric filtering, binding-energy ranking, and visual inspection were combined to reduce millions of docking poses to a small number of experimentally testable candidates. This highlights that, for PET hydrolase engineering, a favorable docking score alone is insufficient; the orientation of the substrate relative to catalytic residues, particularly S160, is also critical for identifying potentially productive enzyme variants.

Consequently, directly docking a single manually constructed PET structure would introduce a strong dependence on the initial geometry. If the starting structure happened to be folded in an unfavorable direction, for example, the docking calculation might fail to identify binding modes that would have been accessible from a more extended conformer.

This issue is particularly relevant to PET hydrolases because their substrates interact with relatively open protein surfaces rather than conventional deep small-molecule binding pockets. Previous mechanistic work on PET hydrolases has likewise emphasized the importance of polymer conformational flexibility. For example, simulations using a PET hexamer showed that PET binding involves conformational rearrangement of the polymer chain as it approaches and enters the active-site region.

For our computational workflow, PET was represented using a trimeric oligomer containing three repeating units. The PET trimer provided a practical compromise between chemical realism and computational cost. A monomer or very short fragment would capture the local ester chemistry but would provide limited information about polymer-chain curvature, aromatic-ring arrangement, and interactions extending along the substrate-binding surface. In contrast, substantially longer oligomers would introduce many additional rotatable bonds and greatly expand the conformational search space.

The trimer therefore retained several features relevant to polymer recognition: multiple aromatic terephthalate groups, multiple hydrolysable ester linkages, flexible ethylene glycol segments, and sufficient chain length to explore different folded and extended arrangements.

Importantly, the PET trimer was treated as a model of a local polymer segment, rather than as a complete representation of the PET material itself.

This distinction is relevant because PET hydrolysis occurs at a heterogeneous polymer–enzyme interface. Our soluble oligomer model cannot reproduce all properties of a PET surface, such as crystallinity, chain packing, or interfacial accessibility. Instead, it allowed us to investigate the molecular interactions that may occur once an accessible PET segment approaches the enzyme.

Why We Separated Conformer Sampling from MD Parameterization

Separating these stages was an intentional part of our workflow. If PET conformational diversity were explored only after assigning a particular classical force field, the sampled ensemble would necessarily reflect the energetic preferences encoded in that force field. Instead, CREST/GFN2-xTB provided an independent semi-empirical description for the initial exploration of ligand conformational space. The resulting ensemble could then be used to generate structurally diverse starting points before the system entered the Amber molecular dynamics framework.

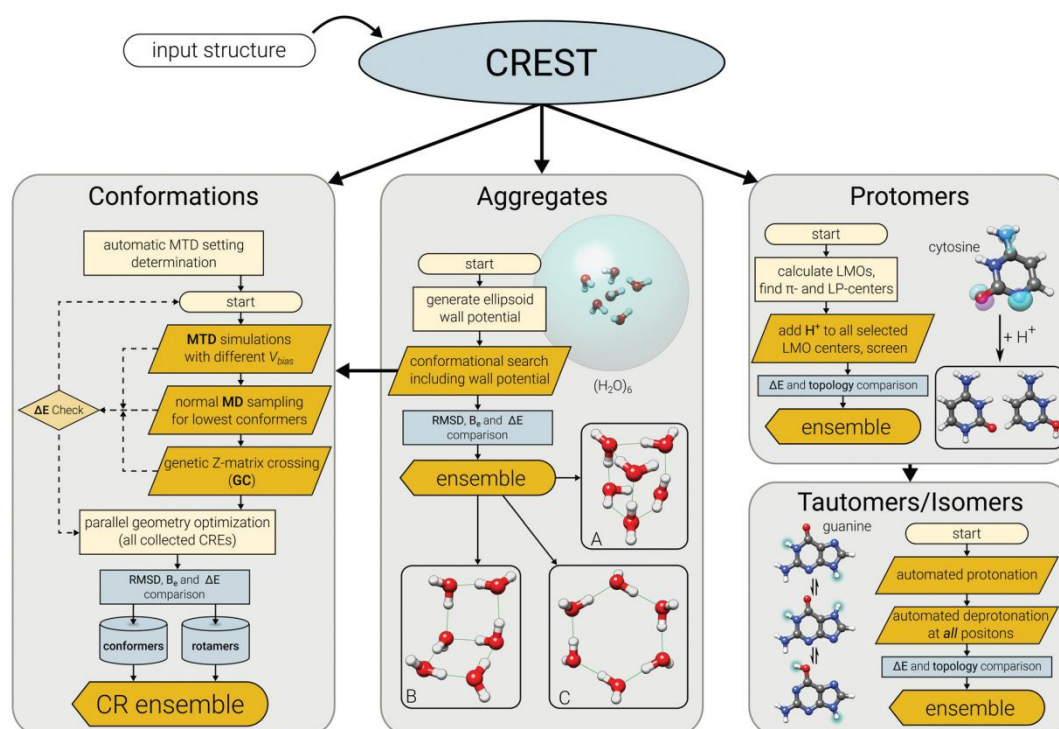


Figure: Overview of the automatized quantum chemical sampling procedures that have been implemented in the CREST code

In this way, conformational exploration and production MD served different purposes:

CREST/GFN2-xTB was used primarily to explore plausible PET oligomer conformations efficiently. RESP/GAFF2 + Amber was subsequently used to describe the selected enzyme–substrate systems consistently during

molecular dynamics simulations. This separation helped prevent our initial structural search from being restricted to a single manually drawn PET geometry and made the transition from ligand preparation to enzyme–substrate simulation more systematic.

I would, however, avoid claiming that this procedure *completely eliminates force-field bias*. Once the selected complexes enter classical MD, their dynamics are still governed by the chosen force field. A more defensible interpretation is that CREST sampling reduces dependence on a single force-field-generated starting conformation before production MD.

Candidate Generation Through Two Design Routes

Overview of Candidate-Generation Strategy

The purpose of candidate generation was not to identify a single “best mutation” directly from one computational score. Instead, we used computation to progressively reduce a very large mutation space into a smaller set of physically plausible and experimentally testable variants.

At the candidate-generation stage, we deliberately separated discovery criteria from mechanistic criteria. A mutation can receive a favorable model probability or docking score without necessarily increasing PET hydrolysis. Conversely, a beneficial mutation may operate through a mechanism that is poorly represented by the original discovery score. For this reason, candidate-generation methods were used primarily to reduce the search space, while later MD, PRS, DFT, and experimental analyses were used to determine whether a mutation actually improved catalysis and why.

This separation was particularly relevant for PET hydrolases because catalytic performance depends on multiple successive processes: substrate recognition, productive positioning, formation of a reaction-ready configuration, and the chemical transformation itself. A candidate-generation algorithm therefore does not need to encode the entire catalytic process to be useful; instead, it needs to enrich the experimentally tested set for variants that are structurally and chemically plausible.

We consequently developed two complementary routes: Route I — Graph-learning-guided design: exploration of local mutational space using a structure-aware GVP-GNN, followed by structural and active-site filtering. Route II — Structure-based active-site design: explicit enumeration of substitutions around the PET-binding region, followed by structural modeling, docking, geometric filtering, and experimental down-selection. This dual strategy reduced dependence on any single design philosophy and also provided an opportunity to compare mutations arising from different parts of the protein and different computational objectives.

Week 5 | 7.01~7.07

Participants: Nianyi Cheng, Xiaofan Luo, Jiayu Wang, Atita Sartsasi, Peilin Zhu, Jiahua Wang, Gao Yishan, Yueyang Du, Haoran Luo, Maoyuan Sun, Yifan Chen

Route I — Graph-Learning-Guided Mutation Design

This dual strategy reduced dependence on any single design philosophy and also provided an opportunity to compare mutations arising from different parts of the protein and different computational objectives.

The first candidate-generation route used a Geometric Vector Perceptron Graph Neural Network (GVP-GNN) to explore mutation possibilities while retaining information about the three-dimensional protein environment.

A protein structure contains information that cannot be represented by amino-acid sequence alone. Residues that are distant in primary sequence can become immediate neighbors after folding, and the effect of a substitution depends strongly on its local packing environment, solvent exposure, backbone geometry, and proximity to functionally important residues. For this reason, we used a structure-aware representation rather than relying exclusively on sequence conservation or manually selected mutation sites.

Conceptually, GVP-based protein models describe the protein as a graph in which residues are represented as nodes and their structural relationships are represented through graph connections. Scalar information can describe properties that do not depend on molecular orientation, whereas vector-valued geometric information allows the model to retain directional features of the three-dimensional structure. This architecture therefore provides a natural framework for combining amino-acid identity with local structural context.

For our engineering workflow, the GVP-GNN was used to evaluate the compatibility of alternative amino acids within the FAST-PETase structural environment. Rather than interpreting the network output as a prediction of enzyme activity, we treated it as a candidate-generation prior: substitutions receiving greater model support were considered more compatible with the learned structural and sequence context and were therefore prioritized for subsequent filtering.

The project Report records that the optimized GVP-GNN explored local mutational space while retaining three-dimensional structural information and produced residue-level amino-acid probabilities that were subsequently combined with structural filtering.

Mutation Generation and Probability Ranking

For each residue considered by the graph-learning model, alternative amino-acid identities were evaluated according to the probability distribution produced by the network. These probabilities provided a systematic way to

rank substitutions rather than selecting mutations solely from visual inspection or chemical intuition.

The ranking was interpreted primarily as a measure of model-supported compatibility with the surrounding protein environment. A high-ranking amino acid therefore indicated that the substitution was considered plausible within the learned sequence–structure landscape; it did not imply that the mutation would necessarily increase catalytic activity.

This distinction was important because several different properties can contribute to a high model score. A substitution may preserve local packing, maintain favorable backbone compatibility, or resemble sequence–structure patterns learned during model training without directly affecting PET hydrolysis. Conversely, catalytically valuable substitutions may act through relatively subtle changes in substrate positioning or conformational dynamics that are not directly encoded by the ranking score.

We therefore used probability ranking as the first reduction of mutation space rather than as the final selection criterion.

In practice, candidate substitutions were organized according to their residue-level probabilities and then mapped back onto the FAST-PETase three-dimensional structure. This enabled sequence-space predictions to be interpreted in structural context before variants were committed to more expensive modeling or experimental testing.

Structural and Active-Site Filtering

Probability ranking alone was insufficient because the engineering objective was specifically to modify PET-hydrolase function. We therefore introduced an additional structural filtering stage to prioritize candidates located in regions with plausible connections to substrate recognition or catalysis.

Candidate residues were mapped onto the FAST-PETase structure and evaluated according to their spatial relationship with the catalytic pocket and the modeled PET-derived substrate. The project Report specifically records that substitutions were filtered according to their proximity to the catalytic pocket or docked PET-derived substrate.

This filtering stage served several purposes. First, it removed candidates that were predicted to be sequence-compatible but were located in structural regions with little obvious connection to the catalytic problem being studied. Second, it retained both direct active-site residues and second-shell residues, allowing the search to extend beyond residues that physically contact the reacting ester group. This was important because second-shell mutations can influence catalysis indirectly by altering packing, hydration, interaction networks, or the conformational ensemble of neighboring active-site residues.

Following the graph-learning and structural-filtering stages, Y87Q emerged as a leading FAST-PETase candidate and was advanced for experimental validation and mechanistic analysis.

Importantly, the rationale at this stage was candidate prioritization rather than mechanistic assignment. The graph-learning model did not establish why Y87Q might improve PET hydrolysis. Instead, it identified Y87Q as a sufficiently promising and structurally plausible substitution to justify experimental testing.

The mechanistic consequences of Y87Q—including preservation of the global fold, local second-shell reorganization, changes in reaction-ready conformational sampling, and subsequent energetic analysis—were evaluated only in the later MD, PRS, and quantum-chemical stages.

Route II — Structure-Based Active-Site Design

Week 6 | 7.08~7.15

Participants: Nianyi Cheng, Xiaofan Luo, Jiayu Wang, Atita Sartsasi, Peilin Zhu, Gao Yishan, Flora Da, Ziyu Zhong, Yifan Chen, Chenming Jin, Ratchakrit Chumchuwat, Bohang Gan

Rationale for Structure-Based Active-Site Design

The second design route used an explicit structure-based strategy to explore substitutions in and around the PET-binding region.

Whereas the graph-learning route searched for substitutions supported by learned sequence–structure relationships, the structure-based route began from a direct mechanistic hypothesis: residues surrounding the bound PET-derived substrate could be modified to alter substrate positioning, local packing, steric complementarity, and the interaction environment of the reactive ester.

This route therefore intentionally concentrated mutational exploration near the enzyme–substrate interface.

The objective was not to maximize substrate-binding affinity alone. PET hydrolysis requires the reacting ester to adopt a geometry compatible with nucleophilic attack by Ser160, and a strongly bound but incorrectly oriented substrate can remain catalytically unproductive. Candidate generation was consequently followed by structural modeling and catalytic-geometry-aware docking rather than ranking variants exclusively by predicted binding energy.

Definition of the Active-Site Design Region

The second design route used an explicit structure-based strategy to explore substitutions in and around the PET-binding region.

Whereas the graph-learning route searched for substitutions supported by learned sequence–structure relationships, the structure-based route began

from a direct mechanistic hypothesis: residues surrounding the bound PET-derived substrate could be modified to alter substrate positioning, local packing, steric complementarity, and the interaction environment of the reactive ester. This route therefore intentionally concentrated mutational exploration near the enzyme–substrate interface.

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Systematic Mutation Enumeration

The tentative amino-acid substitutions were systematically enumerated at the selected design positions to generate a library of candidate FAST-PETase variants.

This strategy allowed the mutation space surrounding the active site to be explored more broadly than would be possible using manual residue-by-residue intuition alone. At the same time, restricting the search to a defined structural region kept the computational problem tractable and preserved a direct mechanistic connection between the mutated residue and the enzyme–substrate interface.

Structural Modeling and Preliminary Evaluation

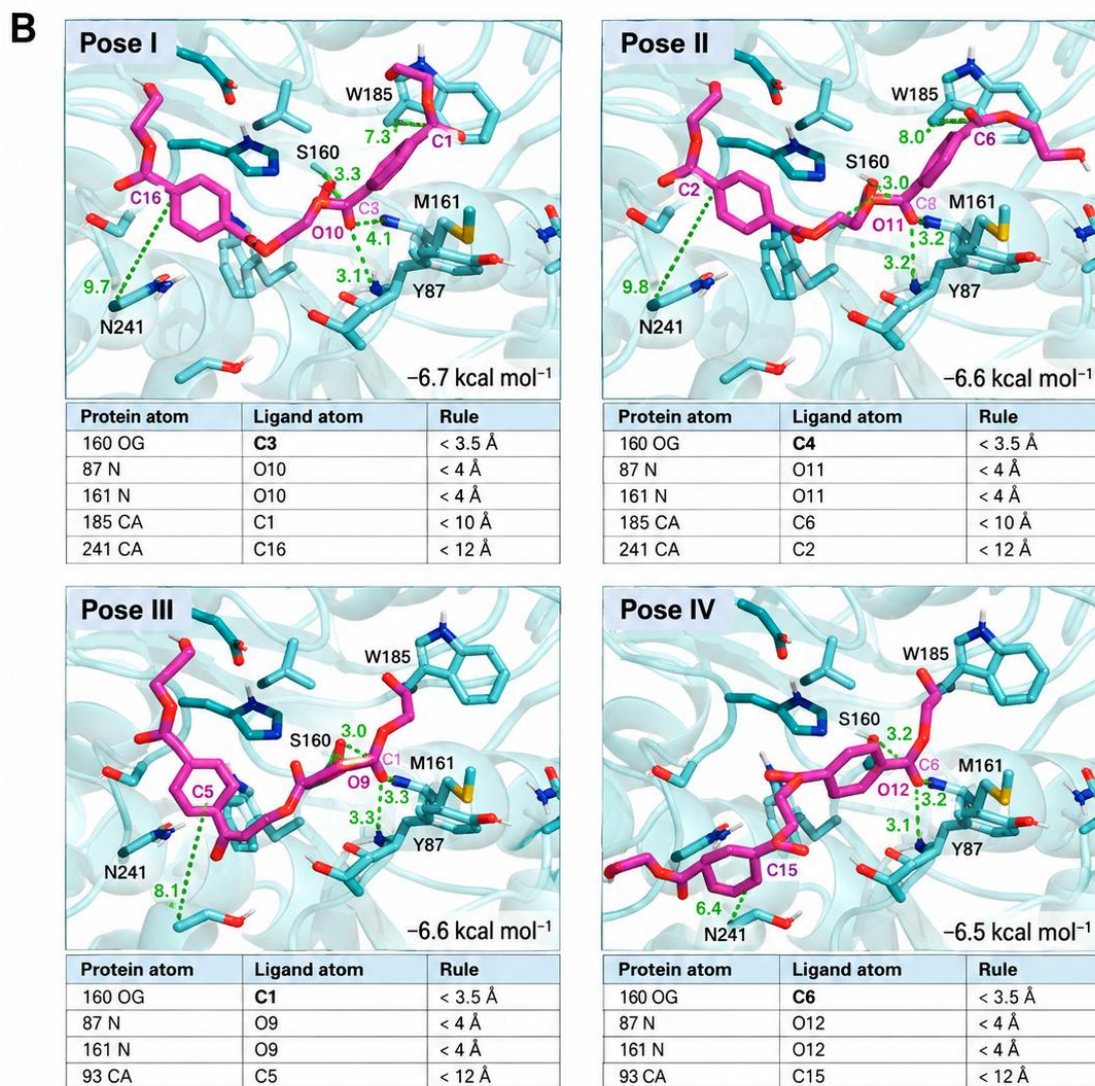
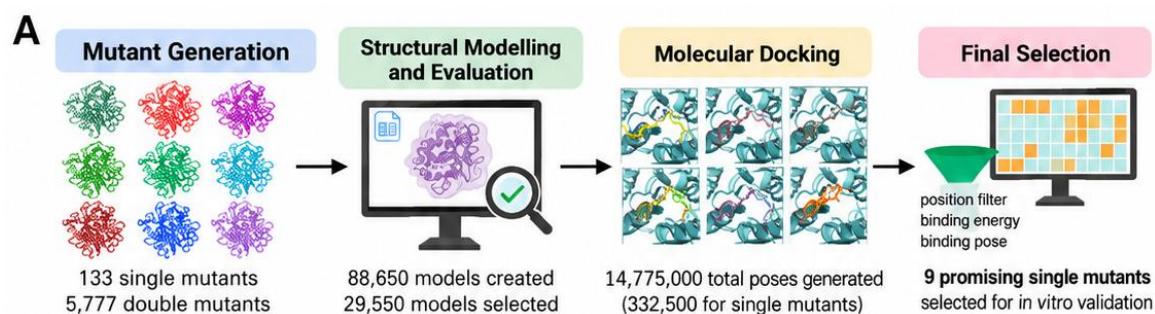
Each enumerated sequence was converted into a three-dimensional structural model before docking. Structural modeling provided an important intermediate quality-control step because not every sequence-level mutation produces a physically reasonable local structure. Depending on side-chain size, polarity, and conformation, a substitution may introduce severe steric overlap, unrealistic packing, or local rearrangements incompatible with the starting enzyme architecture.

The generated models were therefore evaluated before proceeding to ligand docking. The purpose of this stage was not to predict catalytic activity but to remove obviously problematic structural models and ensure that downstream docking was performed on geometrically plausible enzyme variants. large-scale workflow shown in the current Notebook generated tens of thousands of variant structural models before the candidate pool was reduced for the docking stage

Week 7 | 7.16~7.23

Participants: Nianyi Cheng, Xiaofan Luo, Jiayu Wang, Atita Sartsasi, Haoran Luo, Maoyuan Sun, Ziyu Zhong, Xinglin Zhang, Jiahua Wang, Chenming Jin, Gao Yishan, Yifan Chen, Peilin Zhu

Final Selection



Unlike a relatively rigid small-molecule ligand, a PET oligomer contains multiple rotatable bonds connecting aromatic terephthalate groups and ethylene glycol linkers. Rotation around these bonds allows the polymer fragment to adopt substantially different overall geometries, ranging from relatively extended structures to bent or folded conformations.

Molecular Dynamics System Preparation and Production Protocol

Why Molecular Dynamics Is Performed in Stages

A newly constructed solvated system is not immediately suitable for production MD. Even when the original protein structure is experimentally determined or computationally predicted, adding hydrogens, a ligand, solvent molecules, and ions can introduce unfavorable contacts and local geometric strain.

For this reason, we learned to treat MD preparation as a staged relaxation process consisting of energy minimization, heating, equilibration, and production sampling.

Energy minimization is performed first to remove severe steric clashes and relax unfavorable local interactions introduced during system construction. Rather than attempting to reproduce physical molecular motion, minimization moves the system toward a nearby lower-energy configuration.

We focused on establishing the theoretical and practical foundations required for molecular dynamics (MD) simulations of our PET-hydrolase systems. Our main objectives were to understand the roles of force-field parameterization, solvation, energy minimization, heating, equilibration, and production MD, and to translate these concepts into a reproducible AMBER workflow that could later be applied consistently to the wild-type enzyme and designed mutants.

In parallel, we performed quality-control procedures for our wet-lab workflow. PCR products were purified in preparation for downstream cloning and sequencing. Particular attention was given to preventing low-quality experimental or computational inputs from propagating into later stages of the project.

Together, these activities established an important principle for the project: reliable downstream conclusions depend on carefully validated starting materials, whether those materials are DNA fragments used for cloning or molecular structures used for simulation.

Molecular dynamics simulations describe the time-dependent motion of atoms by treating them as classical particles whose interactions are governed by a molecular-mechanics force field. For our project, MD provides a way to

investigate how mutations may alter the conformational behavior of PET hydrolases and their interactions with PET-derived substrates.

Before meaningful trajectories can be generated, however, the molecular system must be carefully parameterized and prepared.

Our enzyme–PET systems contain two chemically different components: a protein enzyme and a PET-derived ligand. These components therefore require compatible but distinct parameterization strategies. We adopted ff19SB for the protein and GAFF2 for the PET-derived ligand. Ligand partial atomic charges were represented using RESP-derived charges, allowing the PET model to be incorporated into the AMBER force-field framework.

The resulting enzyme–PET complex was placed in an explicit TIP3P water box with a minimum solvent shell of approximately 10 Å around the solute. Counterions were then introduced to neutralize the system. This generated the solvated and electrically neutral starting system required for subsequent minimization, equilibration, and production simulations.

An important lesson from this stage was that MD reliability depends strongly on the quality of the starting structure and its parameterization. Missing atoms, unrealistic protonation states, steric clashes, inconsistent ligand parameters, or incorrect system charge can all introduce artifacts that may persist throughout a long simulation. We therefore began treating structure preparation as a formal quality-control checkpoint rather than simply a preliminary step before running MD.

Training Example: AMBER Topology and Coordinate Preparation

The following tleap workflow was used as a training template for understanding how the protein, PET-derived ligand, solvent, and ions are assembled into a simulation-ready system:

```
# Illustrative tleap setup skeleton
source leaprc.protein.ff19SB
source leaprc.gaff2
protein = loadpdb enzyme.pdb
ligand = loadmol2 pet_trimer_resp.mol2
loadamberparams pet_trimer.frcmod
complex = combine {protein ligand}
solvatebox complex TIP3PBOX 10.0
addionsrand complex Na+ 0

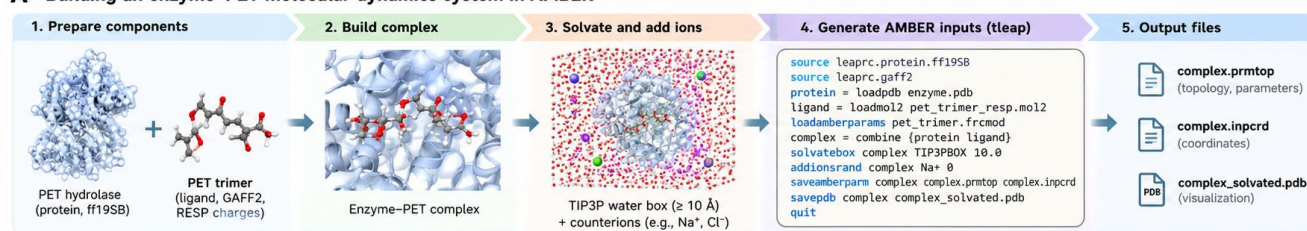
saveamberparm complex complex.prmtop complex.inpcrd
savepdb complex complex_solvated.pdb

quit
```

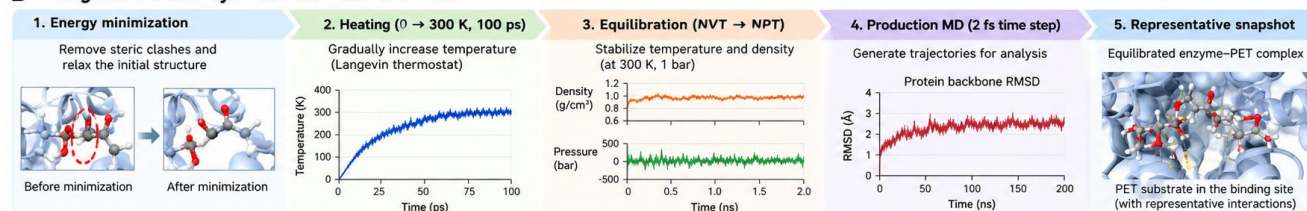
This exercise helped us understand that the topology (prmtop) and coordinate (inpcrd) files are not merely software-specific outputs. Together, they define the chemical identity, connectivity, force-field parameters, and initial geometry of the molecular system that will be propagated during MD.

Before proceeding to simulation, we therefore checked the generated structures for obvious structural abnormalities, inappropriate ligand placement, missing components, and unexpected system composition.

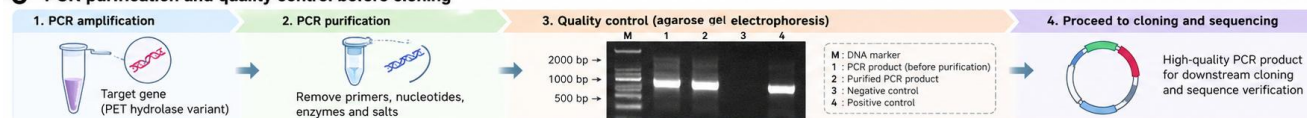
A Building an enzyme–PET molecular dynamics system in AMBER



B Staged molecular dynamics workflow in AMBER



C PCR purification and quality control before cloning



enzyme–PET complex MD simulations

For an enzyme–PET complex, this step is particularly important around the ligand-binding region, where the initial placement of the PET-derived substrate may create short contacts with catalytic-site or neighboring residues.

Following minimization, the system is gradually heated from a low temperature to the desired simulation temperature. In our workflow, the temperature was increased from 0 K to 300 K over 100 ps using Langevin temperature control.

Gradual heating avoids imposing an abrupt thermal perturbation on the newly minimized structure and allows atomic velocities and local conformations to adjust progressively.

Heating was followed by equilibration under NVT and NPT conditions. The purpose of these stages is to allow relevant thermodynamic properties of the solvated system to stabilize before structural data are collected for analysis.

The NVT stage allows the system to adapt at the target temperature while maintaining a constant volume. Subsequent NPT equilibration introduces pressure control, allowing the simulation box and solvent density to adjust under approximately **1 bar** pressure.

During this stage, we learned that reaching the target temperature alone does not demonstrate that a system is equilibrated. Temperature, pressure, density, potential energy, and structural behavior should instead be inspected together before a trajectory is accepted for production analysis.

After equilibration, the system can enter the production stage, during which trajectories are generated for subsequent structural and energetic analyses. A 2 fs time step was used in the planned production workflow.

Importantly, the production trajectory itself is not the final result. It provides an ensemble of molecular conformations from which quantities such as RMSD, RMSF, hydrogen-bond occupancy, enzyme–substrate distances, active-site geometry, and other descriptors can subsequently be calculated.

For our project, these analyses will ultimately allow us to compare the dynamic behavior of the wild-type PET hydrolase and designed mutants under a consistent simulation framework.

Training Example: Staged AMBER Run Sequence

A simplified execution sequence was prepared to understand how the output of each simulation stage becomes the input for the next:

Generic Amber execution pattern

```
pmemd.cuda -O \
-i min1.in \
-o min1.out \
-p complex.prmtop \
-c complex.inpcrd \
-r min1.rst7
```

```
pmemd.cuda -O \
-i heat.in \
-o heat.out \
-p complex.prmtop \
-c min1.rst7 \
-r heat.rst7 \
-x heat.nc
```

```
pmemd.cuda -O \
-i eq_npt.in \
-o eq_npt.out \
-p complex.prmtop \
-c heat.rst7 \
-r eq_npt.rst7 \
-x eq_npt.nc
```

```
pmemd.cuda -O \  
-i prod.in \  
-o prod.out \  
-p complex.prmtop \  
-c eq_npt.rst7 \  
-r prod.rst7 \  
-x prod.nc
```

This staged workflow also highlighted the importance of checkpointing. Each restart file represents a defined state of the system, allowing problems to be traced to a particular preparation or simulation stage rather than requiring the entire workflow to be repeated blindly.

Week 8| 7.24~7.31

Participants: Nianyi Cheng, Xiaofan Luo, Jiayu Wang, Atita Sartsasi, Maoyuan Sun, Chenming Jin, Gao Yishan, Ziyu Zhong, Siqi Ding, Yueyang Du, Haoran Luo, Yifan Chen, Flora Da, Qingxin Ma

CPPTRAJ Trajectory Analysis: From MD Stability to Catalytically Relevant Conformations

Following the production molecular dynamics simulations of wild-type FAST-PETase and the selected engineered variants, the resulting trajectories were analyzed using **CPPTRAJ**, the trajectory-processing and analysis program distributed with the Amber molecular dynamics package. The purpose of this stage was not simply to determine whether the simulated systems remained structurally stable, but to establish whether the enzyme–PET complexes maintained, or repeatedly sampled, conformational states compatible with the proposed catalytic mechanism of PET hydrolysis.

A molecular dynamics trajectory contains a time-dependent ensemble of protein, substrate, solvent, and ion configurations. Although visual inspection can provide an initial qualitative assessment of trajectory behavior, it cannot reliably quantify structural stability, residue-level flexibility, substrate retention, or the frequency with which catalytically productive geometries are sampled. A systematic trajectory-analysis workflow was therefore required to transform the raw MD coordinates into quantitative descriptors that could be compared consistently between wild-type FAST-PETase and the designed mutants.

The CPPTRAJ analysis was organized around two complementary questions. The first was whether the simulated enzyme–PET complexes remained structurally stable throughout the production trajectory. This was evaluated using global and local structural descriptors, including protein backbone RMSD, residue-wise RMSF, radius of gyration, and secondary-structure evolution. The second, and more mechanistically important, question was whether the enzyme maintained geometries compatible with PET

hydrolysis. This required analysis of substrate positioning, enzyme–PET contacts, hydrogen-bond occupancies, catalytic-triad organization, oxyanion-hole interactions, and geometrical parameters associated with nucleophilic attack on the scissile PET ester bond.

Accordingly, the trajectory analysis was designed as a hierarchical workflow progressing from global structural stability, through local active-site organization, to reaction-specific catalytic geometry. This distinction is important because a structurally stable protein is not necessarily catalytically competent. Conversely, local rearrangements within the active site may be mechanistically significant even when the overall protein structure remains essentially unchanged.

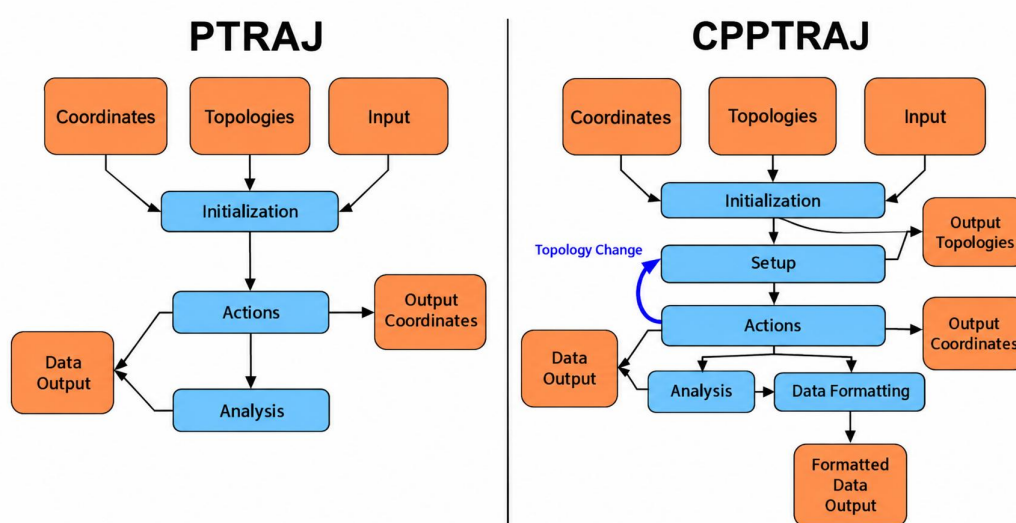


Figure. Overall program flow of PTRAJ and CPPTRAJ. PTRAJ is shown on the left and its C++ successor CPPTRAJ on the right. Orange boxes represent input or output data, whereas blue boxes represent major stages of trajectory processing and analysis. In our workflow, CPPTRAJ was used to convert production MD trajectories into quantitative descriptors for comparing wild-type FAST-PETase and designed mutants.

Protein backbone root-mean-square deviation (RMSD) was calculated to monitor the overall structural evolution of FAST-PETase during the production simulations. RMSD measures the average displacement of selected atoms relative to a reference conformation after optimal structural superposition.

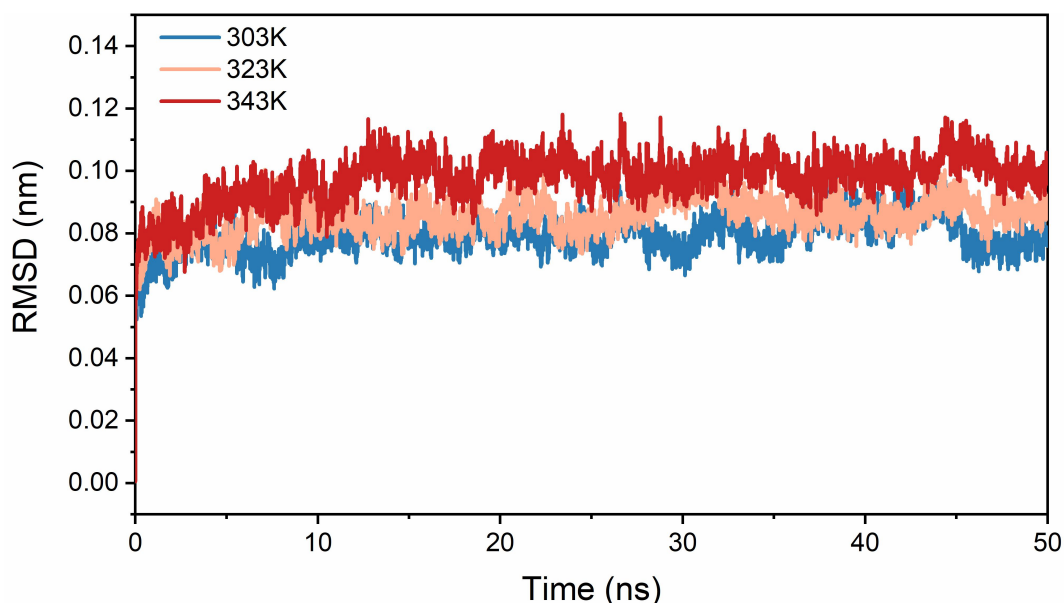


Figure. The changes of root mean square deviation (RMSD) of the C α atoms for FAST-PETase at different temperatures. The data represent the mean values from three parallel simulations.

For each trajectory, backbone atoms were aligned to the reference structure and the corresponding RMSD was calculated as a function of simulation time. A trajectory reaching a relatively stable RMSD regime was interpreted as evidence that the protein had relaxed from the initial structure and was sampling a structurally consistent conformational ensemble.

Establishing a Computational Quality-Control Strategy

One of the most important outcomes of this week's training was recognizing that a successfully completed AMBER job does not necessarily represent a physically meaningful simulation.

We therefore began defining a set of checks that should be completed before accepting an enzyme–PET trajectory for comparative analysis. These include confirming that the expected protein and ligand are present, checking the total charge and ion composition, visually inspecting the solvated structure, examining potential-energy behavior during minimization, and monitoring temperature, pressure, density, and energy during equilibration.

For production trajectories, we plan to examine structural stability using protein backbone RMSD and residue-level flexibility using RMSF. We will also monitor the PET-binding region more specifically, because global protein stability alone cannot determine whether a mutation improves substrate recognition or catalytic geometry.

In later simulations, particular attention will therefore be given to the relative positions of the PET-derived substrate and residues associated with the catalytic center and substrate-binding region. Hydrogen-bond patterns and

relevant interatomic distances will provide additional descriptors for comparing mutant systems with the wild-type enzyme.

Most importantly, we established that all candidate mutants should be evaluated using the same preparation, equilibration, and analysis criteria. Keeping these conditions consistent will make differences between trajectories more interpretable and reduce the risk of confusing methodological variation with mutation-induced effects.

Local Flexibility and Global Structural Stability

To characterize the structural dynamics of wild-type FAST-PETase and the selected mutants, root-mean-square fluctuation (RMSF) and radius of gyration (Rg) were analyzed alongside the global RMSD measurements. RMSF was used to quantify residue-level mobility and to identify local regions whose dynamics were altered by mutation. Particular attention was given to substrate-binding loops, residues surrounding the PET-binding groove, catalytic-site residues, and mutation-adjacent regions. This local analysis was especially important because mutations may modify the flexibility of the substrate-binding environment without producing substantial changes in the overall protein RMSD. Such changes may influence substrate accessibility, active-site organization, and the probability of sampling conformations favorable for catalysis.

The radius of gyration was additionally monitored to evaluate the overall compactness of the enzyme throughout the simulations. Relatively stable Rg values, together with stable backbone RMSD and preserved secondary structure, were taken as evidence that the global protein architecture remained well behaved during MD sampling. Neither RMSF nor Rg was interpreted as a direct predictor of catalytic activity. Instead, these parameters provided complementary information on local flexibility and global structural integrity, establishing the structural context required for subsequent analysis of enzyme–PET interactions and catalytic geometry.

Comparing Wild-Type FAST-PETase and Mutants

To compare wild-type FAST-PETase with the designed mutants, the same trajectory-processing and analysis workflow was applied to every system.

Our comparison therefore did not rely on a single parameter.

Instead, each variant was evaluated across several levels:

Descriptor	Question addressed
Backbone RMSD	Does the overall enzyme remain structurally stable?
RMSF	Which regions become more or less flexible?
Radius of gyration	Does overall protein compactness change?
Secondary structure	Is the protein fold maintained?

Descriptor	Question addressed
H-bond occupancy	Are important enzyme–PET interactions maintained?
Ser O _γ –PET C distance	Is the nucleophile positioned close to the reactive ester?
Attack geometry	Is the substrate oriented appropriately for nucleophilic attack?
PET conformational descriptors	How does the polymer reorganize within the binding groove?

This multi-descriptor strategy prevented us from overinterpreting any single metric.

For example, a mutant with lower backbone RMSD would not automatically be ranked as superior if PET remained in a non-productive binding orientation. Conversely, a mutant displaying somewhat greater loop flexibility could still be mechanistically interesting if this flexibility increased the population of reaction-ready substrate configurations.

Where replicate simulations were available, individual trajectories were retained separately rather than concatenated immediately into a single apparent trajectory. This allowed us to distinguish reproducible behavior from features arising in only one simulation.

Processed numerical data were exported in standardized formats for subsequent plotting and statistical comparison. This workflow ensured that figures shown in the final project could be traced back to a specific trajectory, CPPTraj command, and analysis definition.

MD tells us which conformations the enzyme samples. Trajectory analysis tells us which of those conformations may be reaction-ready. DFT then asks how energetically difficult the chemical transformation is from such a configuration. Together, these analyses provided a multiscale framework for understanding how mutations may influence FAST-PETase catalysis.

Week 9| 8.01~8.07

Participants: Nianyi Cheng, Xiaofan Luo, Jiayu Wang, Atita Sartsasi, Flora Da, Gao Yishan, Yueyang Du, Qingxin Ma, Xinglin Zhang, Haoran Luo, Yifan Chen

AMBER workflow

For our PET hydrolase system, the intended workflow is: Protein–substrate structure → explicit-solvent MD → trajectory sampling → MM/GBSA calculation → energetic ranking → structural inspection. The protein–substrate complex is first subjected to molecular dynamics simulation in explicit solvent. The review emphasizes that explicit-water MD is generally preferable because water molecules can contribute importantly to protein–ligand interactions and implicit-solvent MD may generate less reliable conformational ensembles.

Snapshots are then extracted from the equilibrated MD trajectory. During the MM/GBSA calculation, explicit solvent molecules and ions are removed, and the solvation contribution is recalculated using the implicit GB model. The

individual energy components are averaged over multiple MD snapshots to obtain the final estimate of the binding free energy.

For most protein–ligand systems, the single-trajectory approach is attractive: the complex trajectory is used to generate the complex, receptor, and ligand coordinates for each snapshot. This approach often reduces statistical noise because many internal-energy terms cancel between the complex and its separated components. However, it assumes that ligand binding does not cause very large conformational changes in the receptor or ligand.

3. How AMBER Analyzes MD Trajectories

AMBER

AMBER (Assisted Model Building with Energy Refinement) is a widely used software package for molecular dynamics simulations of biomolecules. It provides a rich set of analysis tools (**cpptraj**) to analyze MD trajectories.

Workflow for trajectory analysis with AMBER (cpptraj):

- 1 Load the trajectory and topology files
- 2 Perform analysis commands
- 3 Output results (plots, data files)

Example cpptraj input file (analyze.in)

```
cpptraj
parm complex.prmtop
trajin md.nc

# RMSD
rmsd :1-300@CA out rmsd.dat

# RMSF
atomicfluct out rmsf.dat byres

# Radius of gyration
gyrate out rg.dat

# Hydrogen bonds
hbond out hbonds.dat

# Interaction energies
autoimage
pairdist :LIG@* :PROT@* out mindist.dat

run
```

Figure. AMBER/CPPTRAJ learning workflow used to understand how topology and trajectory files are loaded and converted into RMSD, RMSF, radius-of-gyration, hydrogen-bond, and interaction-distance outputs. The code shown is a training example; production masks and filenames must match the actual system.

The later production workflow used five independent MD replicas per enzyme. The first 20 ns were treated as additional relaxation, and the 20–100 ns interval was used for standardized cross-system comparison; trajectories that retained a catalytically relevant bound substrate were extended to 1 μ s. Importantly, replica-level averages—not individual trajectory frames—were treated as independent statistical units. This avoids artificial inflation of sample size from the strong temporal correlation between adjacent MD snapshots.

QM-Cluster Construction and DFT Analysis of PET Acylation

Molecular dynamics and pre-reaction-state analysis describe the conformational ensemble available to an enzyme–substrate complex, but they do not directly describe the electronic rearrangements involved in chemical bond formation and cleavage. We therefore introduced a quantum-chemical layer after identifying and clustering reaction-ready configurations from the MD trajectories.

The central question of this stage was:

Once an enzyme–PET complex reaches a reaction-ready configuration, how energetically accessible is the chemical step, and does this energetic cost differ among FAST-PETase variants?

This question is distinct from asking how frequently a reaction-ready geometry is sampled. MD and PRS analysis address access to catalytically competent configurations, whereas density functional theory (DFT) addresses the energetic cost of progressing from such configurations toward the chemical transition state.

Why Classical MD Is Insufficient for Bond Breaking and Formation

Classical molecular dynamics was used to determine how FAST-PETase and its variants sample enzyme–PET conformations over time. These simulations allowed us to evaluate substrate positioning, active-site geometry, local interactions, hydration, and the population of geometric pre-reaction states.

However, conventional classical force fields use a fixed bonding topology. They can describe atomic motion around a predefined molecular connectivity but do not explicitly describe the redistribution of electron density associated with breaking an ester bond, forming a new Ser–substrate bond, or transferring a proton through the catalytic machinery.

This limitation becomes critical during PET acylation. As Ser160 attacks the scissile ester carbonyl, covalent connectivity changes, the carbonyl center evolves toward tetrahedral geometry, charge is redistributed toward the carbonyl oxygen, and proton transfer is coupled to the catalytic histidine. These events require an electronic-structure description.

We therefore used DFT calculations to examine the chemical step explicitly.

The purpose of the QM analysis was not to replace MD, but to answer the next question in the mechanistic hierarchy:

MD: Does the enzyme reach a reaction-ready configuration?

DFT: Once that configuration is reached, what is the energetic cost of chemical transformation?

This distinction is central to the bottleneck-guided framework because a mutation may improve catalysis either by increasing access to reaction-ready

states, by improving their subsequent chemical productivity, or through contributions from both processes.

Construction of the Active-Site QM Cluster

A major consideration in connecting MD to quantum chemistry was the choice of starting structure.

Selecting one visually attractive MD frame would create a strong and difficult-to-quantify structural bias. PET-derived substrates remain conformationally heterogeneous even within the reaction-ready ensemble, and two configurations satisfying the same basic geometric PRS definition may differ in hydrogen-bond organization, substrate distortion, solvation history, or surrounding side-chain arrangements.

We therefore did not treat one MD snapshot as representative of an entire enzyme variant.

Instead, geometric PRS frames were first identified independently within each MD replica and subsequently clustered according to the local active-site geometry. Four representative PRS configurations were selected from the populated conformational states of each of the five independent MD replicas. This produced: 5 independent MD replicas $\times$ 4 representative PRS structures = 20 QM starting structures per enzyme.

Thus, the final QM analysis was based on an ensemble of PRS-derived structures rather than on a single optimized enzyme–substrate complex. This procedure was applied consistently to the enzyme systems included in the mechanistic comparison.

This design serves two purposes.

First, it preserves part of the conformational heterogeneity observed during MD rather than collapsing the enzyme into one arbitrarily chosen structure.

Second, it allows the calculated activation barrier to be interpreted as a distribution conditioned on reaction-ready configurations, rather than as one deterministic number assigned to each enzyme.

Week 10| 8.08~8.15

Participants: Nianyi Cheng, Xiaofan Luo, Jiayu Wang, Atita Sartsasi, Peilin Zhu, Haoran Luo, Gao Yishan, Siqi Ding, Flora Da, Jiahua Wang, Maoyuan Sun, Xinglin Zhang

Construction of the Active-Site QM Cluster

Performing quantum-mechanical calculations on the complete solvated enzyme system would be computationally prohibitive for the number of conformations required in our comparative analysis. We therefore constructed finite all-QM active-site cluster models from the selected PRS structures.

Each cluster retained the molecular components required to represent the local chemistry of PET acylation. These included: the Ser160–His237–Asp206 catalytic machinery; residues contributing backbone N–H groups to the oxyanion-hole environment; the reactive portion of the PET-derived substrate; and neighboring residues required to preserve the principal local interaction network around the reacting ester.

Truncated peptide or side-chain boundaries were capped consistently, and the same active-site composition and truncation strategy were maintained across corresponding parent and mutant systems wherever structurally equivalent residues were available. The purpose was to ensure that energetic differences were associated with changes in the enzyme variants rather than with inconsistent construction of the quantum models.

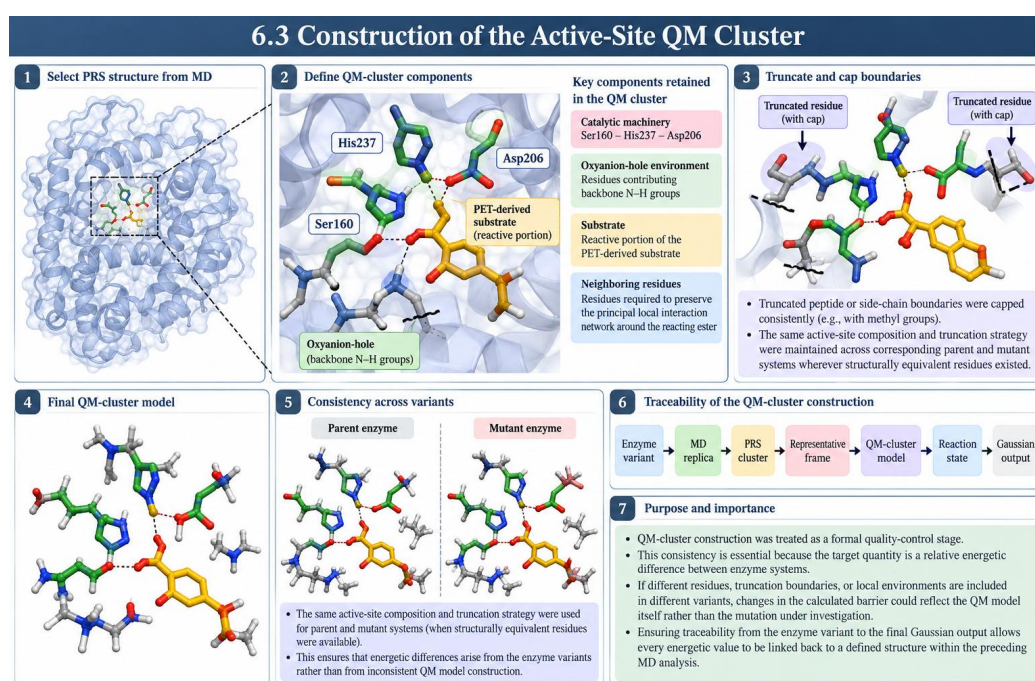


Figure: Construction of the active-Site QM Cluster

This consistency was particularly important because the target quantity was a relative energetic difference between enzyme systems. If different residues, truncation boundaries, or local environments were included in different variants, changes in the calculated barrier could reflect the QM model itself rather than the mutation under investigation. We therefore treated QM-cluster construction as a formal quality-control stage. For every model, the following information was traceable: enzyme variant → MD replica → PRS cluster → representative frame → QM-cluster model → reaction state → Gaussian output

This ensured that every energetic value could be linked back to a defined structure within the preceding MD analysis.

Mechanistic Model of PET Acylation

We focused the quantum-chemical analysis on the acylation step of PET ester hydrolysis.

FAST-PETase follows the general chemistry of a serine hydrolase. In the modeled reaction, the catalytic serine acts as the nucleophile, while His237 participates in proton transfer and Asp206 supports the catalytic histidine. The essential chemical events can be summarized as follows.

First, Ser160 is activated within the catalytic proton-transfer network and its O_y atom approaches the carbonyl carbon of the scissile PET ester.

As nucleophilic attack proceeds, a new interaction develops between Ser160 O_y and the ester carbonyl carbon. The initially planar carbonyl center evolves toward a tetrahedral configuration, while increased negative character develops on the carbonyl oxygen.

The oxyanion-hole region contributes to stabilization of this developing electronic structure.

Accordingly, the modeled reaction cannot be represented by one distance alone. The corresponding energy profile provides a quantitative description of the energetic accessibility of this modeled chemical step. Importantly, our calculations did not suggest that Y87Q or S187Y introduced an alternative chemical pathway. Instead, the comparative question was whether the mutations changed the energetic cost of reaching and stabilizing the same general acylation transition-state region.

Reactant-State Optimization

For each FAST-PETase variant, the initial enzyme–PET geometry was optimized to obtain a reactant-state structure. Frequency calculations were subsequently performed to characterize the optimized stationary point. A valid reactant-state structure was expected to behave as a local minimum on the calculated potential-energy surface.

Particular attention was paid to the geometry of the catalytic center. The PET ester carbonyl needed to remain positioned relative to the catalytic serine in a configuration compatible with nucleophilic attack, while the catalytic hydrogen-bond network also needed to remain chemically reasonable after optimization.

This step was important because comparing transition-state energies against poorly defined or structurally inconsistent reactant states would make the resulting activation barriers difficult to interpret.

Locating the Acylation Transition State

The next step was to identify the transition state associated with nucleophilic attack on the PET ester bond. Transition-state optimization differs fundamentally from ordinary geometry optimization. Whereas a minimum lies at the bottom of a local region of the potential-energy surface, a first-order transition state represents a saddle point separating neighboring regions along a reaction pathway.

A representative Gaussian input template was:

```
%chk=prs_ts.chk
```

```
%nprocshared=16
```

```
%mem=32GB
```

```
#p wb97xd/6-31+G(d) opt=(ts,calcfc,noeigentest) freq
```

```
scrf=(smd,solvent=water)
```

FAST-PETase PET acylation transition-state search

<coordinates>

The opt=ts keyword directs the optimization toward a transition-state structure, while calcfc calculates the initial force constants to assist the optimization. Frequency analysis was performed following optimization to determine the nature of the stationary point. However, convergence of the optimization was not considered sufficient evidence of a valid transition state.

Week 11 | 8.16~8.23

Participants: Nianyi Cheng, Xiaofan Luo, Jiayu Wang, Atita Sartsasi, Haoran Luo, Siqi Ding, Peilin Zhu, Ziyu Zhong, Flora Da, Qingxin Ma, Maoyuan Sun, Xinglin Zhang, Gao Yishan

Comparison of Wild-Type FAST-PETase and Mutants

Once corresponding stationary points had been identified, the same computational protocol was applied consistently to wild-type FAST-PETase and each selected mutant. For each system, the activation barrier of the acylation step was calculated relative to its own optimized reactant state.

This normalization was essential because absolute electronic or Gibbs energies of different structures are not meaningful measures of catalytic performance by themselves. Instead, the relevant quantities are the relative energetic differences between corresponding states along the same modeled reaction pathway. Within the assumptions of the QM-cluster model, a lower acylation barrier therefore indicates a more energetically favorable chemical step, whereas a higher barrier suggests reduced accessibility of the modeled reaction pathway.

Connecting Mutation, Structure, and Activation Energy

The analysis was not limited to identifying which mutant exhibited the lowest activation barrier; an equally important objective was to understand why the barrier changed. Transition-state structures were therefore examined together with their calculated energies, with particular attention to the nucleophilic attack geometry of Ser160, the Ser160–His237–Asp206 catalytic network, oxyanion-hole stabilization, and interactions surrounding the reactive PET ester group.

This structural interpretation connected the MD and DFT components of the workflow. MD simulations describe how frequently the enzyme samples reaction-ready conformations, whereas DFT calculations estimate the energetic cost of progressing through the chemical transformation from such conformations. A mutation may therefore influence catalysis in two distinct but complementary ways: by changing the probability of forming a catalytically competent configuration, or by altering the activation energy of the chemical step itself. Combining MD-derived conformational sampling with QM calculations allowed both effects to be evaluated within a unified mechanistic framework.

Barrier Distributions across PRS-Derived Structures

Because four representative PRS structures were selected from each of five independent MD replicas, each enzyme variant was characterized using 20 PRS-derived QM reaction pathways rather than a single optimized structure. This ensemble-based strategy allowed the distribution of calculated acylation barriers to be examined and reduced the risk that conclusions would be dominated by one unusually favorable or unfavorable conformation.

The 20 QM calculations were treated as nested conformational samples rather than 20 independent simulation replicates, since structures originating from the same MD trajectory are not fully independent. The five MD replicas therefore remained the higher-level sampling units for statistical comparison. Barrier values from PRS structures belonging to the same replica could first be summarized at the replica level, while the complete set of QM results was retained to describe conformational variability and visualize the overall barrier distribution. Using this framework, mean acylation barriers and their associated variability could be compared systematically between wild-type FAST-PETase and the engineered mutants.

Connecting Gaussian to Multiwfn

We also learned to preserve wavefunction information from Gaussian calculations for subsequent analysis using Multiwfn. The optimized reactant, transition-state, and intermediate structures could therefore support

complementary electronic-structure analyses, including IGMH, QTAIM, electrostatic-potential mapping, and orbital-based interpretation.

These analyses may help explain why a particular mutant exhibits a different calculated activation barrier. For example, changes in non-covalent interactions or electron-density distribution around the transition-state region may provide mechanistic context for differences observed in the reaction-energy profiles. However, no individual descriptor was treated as proof of the catalytic mechanism.

Instead, we considered geometry, activation energy, electron density, non-covalent interactions, and orbital characteristics as complementary views of the same reaction process.

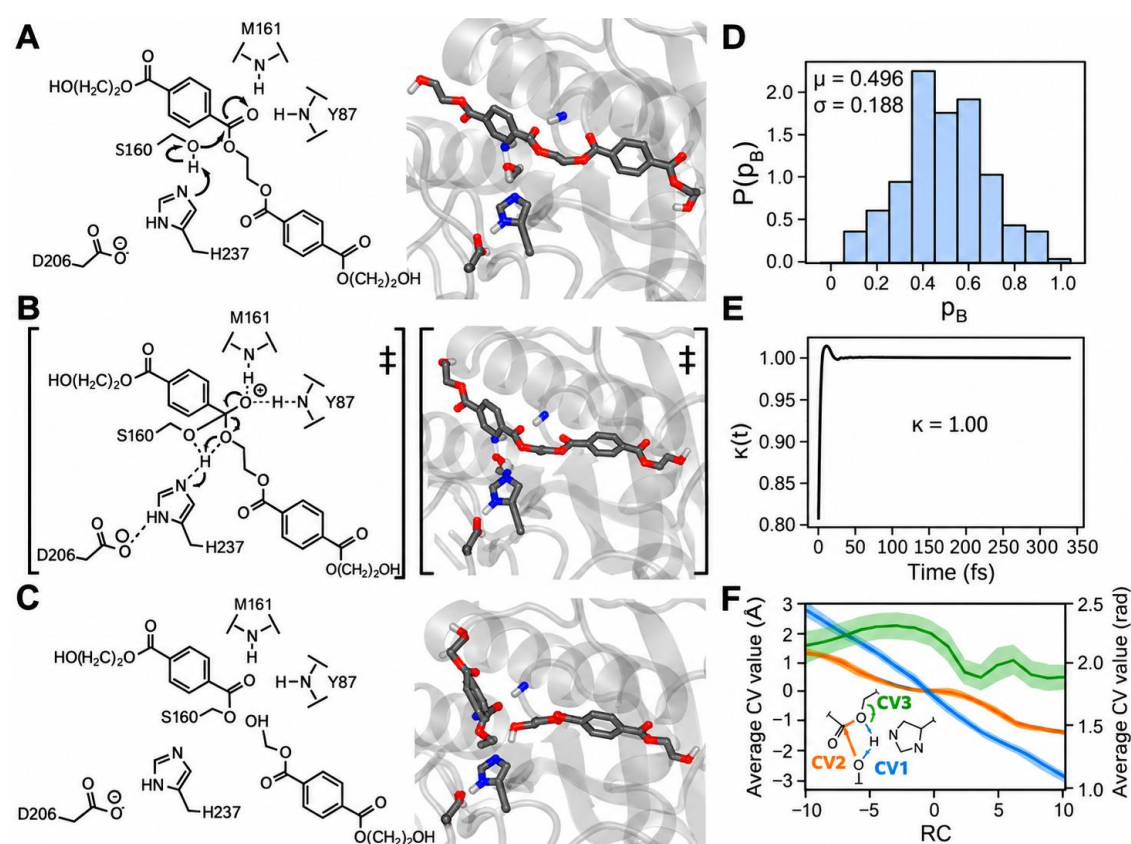


Figure: PETase acylation reaction. A–C Scheme depicting the steps of the acylation reaction with schematics (left) and corresponding model snapshots (right). Michaelis complex, where the catalytic Ser160 and PET ligand are oriented for reaction. TS configuration where Ser160 has attacked the carboxyl carbon of PET, Ser160 has transferred the proton to His237, the carboxyl carbon of PET adopts a tetrahedral geometry, and His237 is midway through the proton transfer.

The acylation reaction coordinate (RC) provides a quantitative description of these coupled structural changes rather than representing only a single

bond distance. The optimized RC contains collective variables describing the differences between distances associated with proton transfer, nucleophilic bond formation, and cleavage of the PET ester bond. Specifically, one component monitors transfer of the Ser160 proton toward the eventual BHET oxygen, while another describes the competition between formation of the Ser160 O–PET carbonyl C bond and rupture of the original PET C–O bond. An additional angular term describes the orientation of the transferred proton relative to the oxygen atom that ultimately accepts it and the adjacent carbon atom. This angle increases slightly from approximately 2.1 to 2.2 radians as the proton is transferred from Ser160 toward His237, and then decreases toward approximately 1.8 radians as proton transfer to the leaving group is completed. Importantly, the bond-forming and bond-breaking coordinates evolve together, further indicating that nucleophilic attack, ester cleavage, and proton shuttling are strongly coupled during acylation. The numerical RC itself is dimensionless and has an arbitrary scale, with negative values corresponding to reactant-like configurations and positive values corresponding to product-like configurations.

Week 12 | 8.24~8.31

Participants: Nianyi Cheng, Xiaofan Luo, Jiayu Wang, Atita Sartsasi, Ratchakrit Chumchuwat, Jiahua Wang, Ziyu Zhong, Yueyang Du, Maoyuan Sun, Yifan Chen, Haoran Luo, Qingxin Ma, Gao Yishan, Xinglin Zhang

Application to PET hydrolase engineering

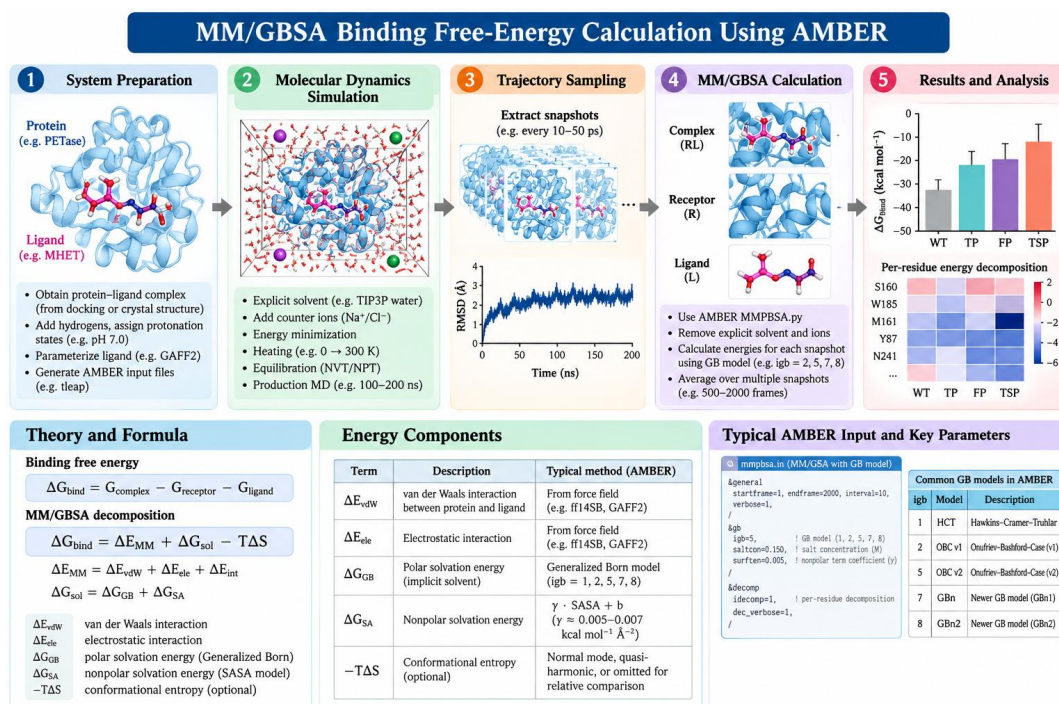
For PET hydrolases, MM/GBSA can complement docking by asking a different question. Docking mainly evaluates whether a PET-derived ligand can adopt a plausible pose in the active site, whereas MM/GBSA estimates how energetically favorable that protein–substrate interaction remains across MD-sampled conformations. In our workflow, we would therefore use MM/GBSA to compare WT and engineered PETase variants and examine contributions such as:

van der Waals interactions → electrostatic interactions → polar solvation → nonpolar solvation → total relative binding energy

Per-residue energy decomposition can further help identify mutations that alter favorable interactions around the PET-binding region.

However, a more favorable MM/GBSA value should not automatically be interpreted as higher catalytic activity. MM/GBSA is an end-point binding method and describes non-covalent association rather than the chemical reaction pathway itself. For PET hydrolysis, a substrate may bind strongly while adopting a geometry that is poorly aligned for nucleophilic attack by the catalytic serine. We would therefore interpret MM/GBSA together with

catalytic-distance analysis, substrate orientation, MD stability, and experimental hydrolysis measurements, rather than using binding energy alone as the criterion for selecting PETase mutants. This follows directly from the end-point nature of the method and from the review's emphasis that predictions depend strongly on sampling and model assumptions.



We studied MM/PBSA and MM/GBSA in AMBER as post-MD approaches for estimating relative protein–substrate binding energetics. Compared with docking scores, MM/GBSA incorporates molecular-mechanics interactions and implicit-solvent contributions and can therefore provide a more detailed energetic comparison of PETase variants. At the same time, we documented that the method is approximate and highly sensitive to conformational sampling, dielectric treatment, force-field parameters, and entropy assumptions. Consequently, MM/GBSA is most useful in our PET hydrolase project as a comparative rescoring and mechanistic-analysis tool, rather than as an absolute measure of catalytic performance. A strongly favorable binding energy still needs to be evaluated together with the geometry of the catalytic complex and subsequent experimental activity.

Learning NCI, IGMH, and QTAIM Analysis for PET Enzyme–Substrate Interactions

As our computational work progressed from molecular visualization and molecular dynamics simulations toward a more detailed interpretation of enzyme–substrate interactions, we began learning real-space electron-density analysis, with particular emphasis on Noncovalent Interaction (NCI) analysis,

the Independent Gradient Model based on Hirshfeld partition (IGMH), and the Quantum Theory of Atoms in Molecules (QTAIM). These approaches provide complementary descriptions of molecular interactions based on electron density and its derivatives, transforming otherwise abstract electronic information into spatially resolved representations. The literature emphasizes that real-space wavefunction analyses can provide information complementary to conventional geometry optimization and energetic calculations, particularly by revealing where interactions occur and how their characteristics vary across a molecular system.

This training was particularly relevant to our project because the catalytic degradation of polyethylene terephthalate (PET) depends not only on the overall binding of the polymer to the enzyme, but also on the local molecular environment formed between the PET substrate and residues surrounding the catalytic region. Before hydrolysis can occur, the PET chain must adopt a productive binding configuration in which the target ester group is appropriately positioned relative to the catalytic machinery. Hydrogen bonding, van der Waals contacts, steric effects, and interactions involving the aromatic terephthalate moiety may collectively influence substrate recognition and stabilization. Therefore, rather than describing enzyme–substrate recognition using only individual interatomic distances obtained from MD trajectories, we learned how real-space analyses could provide a more spatially continuous picture of the interaction environment.

We first studied NCI analysis as an intuitive approach for identifying regions associated with noncovalent interactions. NCI uses the reduced density gradient (RDG) together with the electron-density-based quantity $\text{sign}(\lambda_2) \rho$. In conventional NCI visualization, negative $\text{sign}(\lambda_2)$ values are associated with attractive regions, whereas positive values can indicate repulsive or sterically crowded regions; the magnitude of the electron density provides qualitative information about interaction strength. Importantly, the literature cautions that $\text{sign}(\lambda_2) \rho$ should not be overinterpreted as a direct measure of interaction energy.

For our PET-degradation project, this framework provided a useful conceptual extension beyond simple distance measurements. For example, a conventional MD analysis may show that an oxygen atom of a PET ester group remains close to a polar residue, but distance alone does not fully describe the spatial character of the surrounding interaction. An NCI surface can instead indicate whether an extended attractive region is present between the substrate and its surrounding residues, while simultaneously revealing weak dispersive contacts and regions of steric crowding. The literature illustrates this distinction particularly clearly: attractive hydrogen-bonding regions, weak π -related or van der Waals contacts, and sterically repulsive regions can occupy visibly different regions of an NCI map.

This perspective is valuable when considering PET because its repeating unit contains both polar ester groups and aromatic terephthalate rings. Consequently, productive substrate binding may arise from several weak interactions acting cooperatively rather than from a single dominant contact. In our later computational analysis, NCI-type visualization could therefore be applied to representative enzyme–PET configurations selected from molecular dynamics simulations to investigate how mutations reshape the local noncovalent environment surrounding the bound substrate.

We next studied IGMH, which was particularly attractive for enzyme–substrate analysis because it allows interactions to be examined in a fragment-based manner. In IGMH, the actual molecular electron density is partitioned using the Hirshfeld scheme, and interfragment and intrafragment interactions can be considered separately. This means that, for a representative PET–enzyme model, the PET substrate and selected active-site residues could be defined as different fragments so that the interaction between them can be examined more directly.

The advantage of this approach became clearer from the examples discussed in the literature. For a capped phenylalanine trimer, fragment definitions allowed IGMH to focus specifically on intermolecular interactions and visualize both van der Waals and hydrogen-bonding regions. Compared with conventional NCI visualization, the resulting interaction regions were less scattered and easier to associate with the fragments of interest.

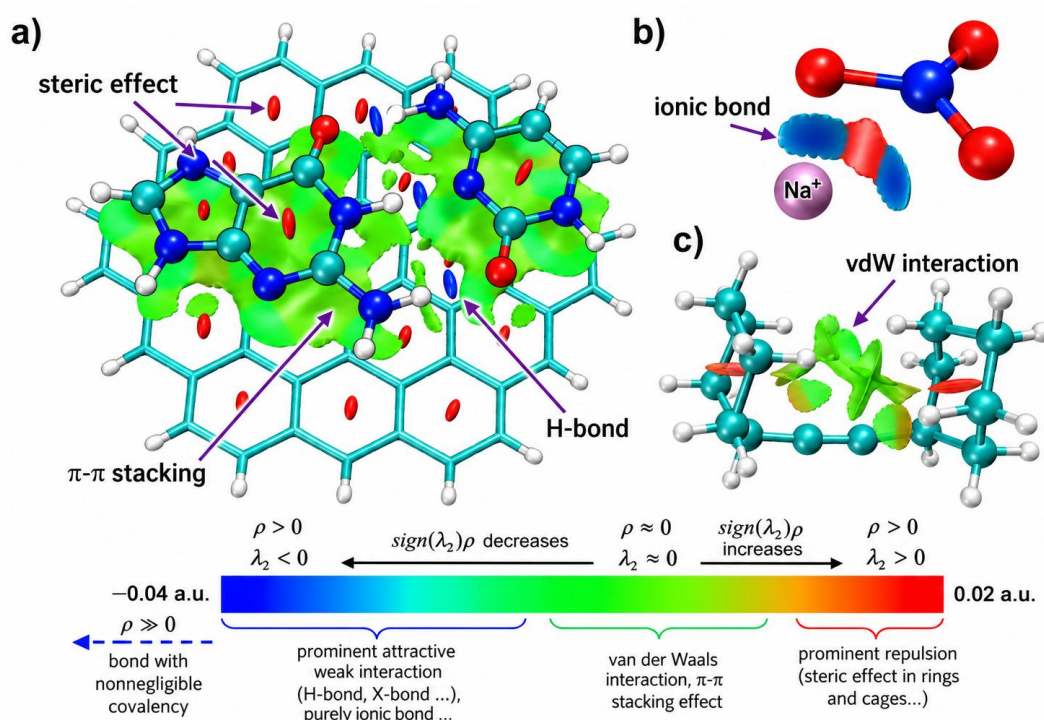


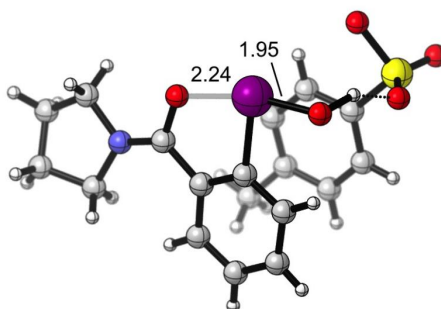
Figure. NCI maps of representative systems highlighting noncovalent interactions.

Learning CYLview for Molecular Visualization

During this week, we became familiar with CYLview as a molecular visualization tool for inspecting and presenting three-dimensional structures generated during our computational studies. We practiced visualizing and adjusting molecular geometries, highlighting atoms and functional groups of interest, and preparing clear structural representations. In the context of PET biodegradation, CYLview provided a convenient way to display PET-derived substrates and hydrolysis products, such as MHET and TPA, as well as representative structures extracted from our calculations. We were particularly interested in using CYLview to illustrate the catalytic mechanism of PET hydrolysis. By comparing structures corresponding to different stages of the proposed reaction pathway, we could more clearly present changes around the ester bond undergoing hydrolysis and visualize the relative arrangement of the substrate and catalytically relevant groups. When combined with structures obtained from molecular dynamics and quantum-chemical calculations, these visualizations helped us communicate how PET-derived substrates may progress through different catalytic configurations and reaction intermediates.

CYLview

User Manual



Finally, we learned to optimize molecular orientation, representation, labeling, and rendering for publication-quality figures. Importantly, we treated CYLview primarily as a visualization and communication tool, while the underlying structural, energetic, and mechanistic conclusions were derived from the corresponding molecular simulations and quantum-chemical calculations

Multwfn Wavefunction Analysis

To investigate the electronic origin of these energetic differences, wavefunctions obtained from the optimized DFT structures were further analyzed using Multwfn. We focused particularly on the Independent Gradient Model based on Hirshfeld partitioning (IGMH) and Quantum Theory of Atoms in Molecules (QTAIM) analyses. These approaches provided complementary descriptions of the noncovalent interaction network surrounding the reactive PET ester and catalytic residues. IGMH offered an intuitive real-space visualization of interaction regions, whereas QTAIM allowed selected interactions to be examined through the topology of the calculated electron density. By applying the same analyses to corresponding WT and mutant structures, we could determine whether mutations reorganized the electronic environment of the active site as the system approached the transition state.

The Multwfn analysis allowed us to move beyond a purely structural description of the mutants. Two enzyme variants can display similar catalytic distances in MD yet generate different electronic environments as the reaction approaches the transition state. Conversely, an interaction that appears geometrically minor may contribute to organizing the substrate, reducing unfavorable steric contacts, or stabilizing the developing charge distribution of the transition-state region. We therefore compared WT and mutant systems across the same sequence of structures and asked whether favorable interaction regions were strengthened, unfavorable contacts were relieved, or the local interaction network became better organized in the mutant transition state.

These observations provided a mechanistic bridge between the MD and DFT results. MD described how mutations reshaped the conformational ensemble and the probability of sampling reaction-ready geometries; DFT quantified the energetic cost of the chemical step; and Multwfn helped reveal how the underlying electron-density and noncovalent-interaction patterns changed along that step. This multiscale interpretation was important because a reduced activation barrier cannot necessarily be attributed to one newly formed hydrogen bond or one altered residue. Instead, catalytic improvement may emerge from the collective reorganization of several individually modest interactions within the active site.

Recombinant Protein Expression and Purification

Following computational screening, molecular docking, and molecular dynamics analysis, a subset of FAST-PETase mutations was selected for experimental validation. The purpose of the wet-lab workflow was to translate these *in silico* predictions into physical enzyme variants and determine

whether the structural and dynamic changes predicted computationally could ultimately produce measurable differences in enzyme performance.

To make comparisons between variants as meaningful as possible, wild-type FAST-PETase and the selected mutants were processed using the same general experimental workflow. Wherever possible, culture conditions, induction strategy, cell disruption, purification, and downstream sample handling were kept consistent between variants. This was important because differences in protein yield, solubility, or apparent activity can arise from experimental conditions independently of the designed mutation.

Our experimental workflow therefore followed a continuous Build–Test sequence:

Mutation design → Mutant plasmid construction → Sequence verification → Transformation → Recombinant expression → Cell harvest and lysis → Ni²⁺ affinity purification → Protein quality assessment → Enzymatic characterization

In this way, each computationally selected mutation could be traced from a predicted structural hypothesis to a sequence-confirmed construct and, ultimately, to a purified protein suitable for experimental testing.

Week 13| 9.01~9.07

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Transformation into E. coli BL21(DE3)

The coding sequence of FAST-PETase was prepared in the pET-22b(+) expression vector, using the NdeI and XhoI restriction sites. A C-terminal 6×His tag was retained in the expression construct to facilitate subsequent purification by immobilized metal affinity chromatography (IMAC).

The wild-type FAST-PETase construct served as the reference for the experimental comparison. Candidate mutations identified during the computational design stage were introduced into the FAST-PETase coding sequence by site-directed mutagenesis, generating the corresponding mutant expression plasmids.

Because a computationally designed mutation does not become an experimentally testable variant until the corresponding genetic construct has been physically verified, plasmid construction was followed by sequence confirmation. Candidate clones were screened, and the relevant regions of the FAST-PETase coding sequence were examined by DNA sequencing.

A mutant construct was considered successfully generated only when the sequencing result confirmed the intended nucleotide substitution and

corresponding amino-acid change. The verified sequence was also inspected for unintended changes within the sequenced region before the construct was carried forward into protein expression.

This sequence-verification step formed an important checkpoint in our engineering cycle. Computational modelling represented the Design stage, whereas the sequence-confirmed expression plasmid represented the first physical realization of that design—the Build stage.

By introducing this checkpoint before protein production, we reduced the possibility that differences observed later during expression, purification, or activity measurements could be incorrectly attributed to the intended mutation when the underlying construct was not correct.

Transformation into *E. coli* BL21(DE3)

Sequence-verified plasmids encoding wild-type FAST-PETase or the designed variants were introduced into chemically competent *Escherichia coli* BL21(DE3) cells for recombinant protein production. For each transformation, 2 μ L of plasmid solution was gently mixed with thawed competent cells and incubated on ice for 30 min. The cells were subsequently heat shocked at 42 °C for 90 s and immediately returned to ice for 2 min. Following heat shock, 900 μ L of LB medium was added to the cells, which were then allowed to recover at 37 °C and 200 rpm for 1 h. This recovery period allowed the transformed cells to resume growth and express the antibiotic-resistance marker before selective plating. The recovered cultures were spread onto LB agar plates containing 100 μ g/mL ampicillin and incubated overnight. Colony formation on the selective plates provided the first indication that cells carrying the expression plasmid had been successfully recovered. The same transformation procedure was used for the wild-type construct and designed variants so that subsequent protein-production experiments began from comparable expression hosts.

Starter Culture and Scale-Up

For each construct, a single, well-isolated colony was selected from the transformation plate and inoculated into 5 mL LB medium containing ampicillin to establish a starter culture.

Starting from an isolated colony was important because it reduced the likelihood that a mixed bacterial population would enter the protein-expression workflow. It also allowed each expression experiment to be traced back to an individual transformed clone.

The starter cultures were incubated at 37 °C with shaking until robust bacterial growth was observed. These cultures were subsequently used to inoculate larger-volume expression cultures.

This seemingly simple step was important for experimental reproducibility. Since our objective was to compare FAST-PETase variants rather than differences in bacterial growth history, the same general starter-culture procedure was followed for each construct.

Recombinant Expression of FAST-PETase Variants

For the recorded small-scale protein-production workflow, 2 mL of starter culture was transferred into 200 mL LB medium containing ampicillin. The cultures were grown at 37 °C with shaking, and bacterial growth was monitored by measuring the optical density at 600 nm (OD600). Expression was initiated when the cultures reached the mid-logarithmic growth phase, corresponding to an OD600 of approximately 0.6–0.8.

Monitoring OD600 before induction provided a reproducible physiological reference point for initiating protein expression. This was particularly important for comparisons between variants: if one culture were induced during early growth and another near stationary phase, subsequent differences in recombinant protein yield could reflect differences in cellular state rather than properties of the FAST-PETase variants themselves.

Protein expression was therefore induced only after cultures had reached a comparable growth phase. The same induction strategy was applied to wild-type FAST-PETase and the mutant constructs included within each experimental comparison.

Following induction, the cultures were incubated under the corresponding recorded expression conditions to allow recombinant FAST-PETase to accumulate before cell harvest.

For the final Notebook, the inducer concentration, induction temperature, shaking speed, and induction duration should be reported exactly as recorded for each experimental batch. These parameters are particularly important because changing induction temperature or induction strength can substantially alter recombinant protein yield and the proportion of protein recovered in the soluble fraction.

IPTG Induction

When the expression cultures reached an OD600 of approximately 0.6–0.8, corresponding to the mid-logarithmic growth phase, recombinant protein expression was induced by adding isopropyl β -D-1-thiogalactopyranoside (IPTG) to a final concentration of 0.1 mM. The cultures were then transferred to 20 °C and incubated with shaking at 200 rpm for approximately 24 h. These lower-temperature induction conditions were selected to allow recombinant FAST-PETase to accumulate while reducing the risk of rapid overexpression leading to misfolding, aggregation, or loss of soluble protein.

The same induction strategy was applied to the corresponding wild-type and mutant constructs within each experimental comparison. Maintaining a consistent induction point, IPTG concentration, temperature, and shaking condition was important because variation in expression conditions can influence total protein yield and the proportion of recombinant protein recovered in the soluble fraction. Thus, differences observed during subsequent purification or catalytic assays were interpreted only after confirming that the compared variants had been produced under matched expression conditions. Following the induction period, the cultures were immediately processed for cell harvest and protein extraction.

Cell Harvest and Lysis

Following recombinant expression, bacterial cells were collected by centrifugation at $8,000 \times g$ for 10 min at 4 °C. The harvested cell pellets were washed twice with buffer and subsequently resuspended in 50 mM Tris-HCl, pH 7.5. Performing the harvesting and resuspension steps at low temperature helped limit protein degradation and maintain the stability of the expressed PET hydrolase before purification.

To facilitate cell disruption, lysozyme was added to the resuspended cells to a final concentration of 1 mg/mL, followed by sonication using alternating 3-s pulses and 5-s pauses at 35% amplitude. Intermittent sonication was used to disrupt the bacterial cells while limiting excessive heat accumulation during the lysis process. After sonication, insoluble cell debris and unbroken cells were removed by centrifugation at $8,000 \times g$ for 1 h at 4 °C. The resulting clarified supernatant contained the soluble recombinant protein and was further passed through a 0.22- μ m filter before loading onto the nickel-affinity column.

The soluble and insoluble fractions were retained where appropriate for SDS-PAGE analysis. This comparison provided an important quality-control step because successful recombinant expression does not necessarily imply that the target protein is recovered in a soluble form. Only after confirming that FAST-PETase or the corresponding mutant was present in the clarified soluble fraction was the sample carried forward into Ni^{2+} -affinity purification.

Week 14| 9.08~9.15

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Ni²⁺ Affinity Purification of FAST-PETase Variants

Following recombinant expression and cell lysis, wild-type FAST-PETase and the designed variants were purified from the clarified soluble fraction using **Ni²⁺-based immobilized metal affinity chromatography (IMAC)**. All constructs contained a C-terminal **6×His tag**, providing a common purification handle and allowing the same general purification strategy to be applied across different variants.

Before sample loading, the nickel-affinity matrix was equilibrated with the corresponding binding buffer. The clarified soluble lysate was then applied to the column. Under the selected buffer conditions, the histidine residues of the C-terminal His tag coordinated with immobilized Ni²⁺ ions on the affinity matrix, allowing recombinant FAST-PETase to remain associated with the column while a substantial proportion of unbound host-cell proteins passed through.

Following sample loading, the column was washed with buffer containing a relatively low concentration of **imidazole**. This step was used to reduce nonspecific background by removing host proteins that interacted weakly with the affinity matrix, while retaining the more strongly bound His-tagged FAST-PETase.

The target protein was subsequently recovered using an elution buffer containing a higher concentration of imidazole. At elevated concentrations, imidazole competes with the histidine side chains for coordination with the immobilized Ni²⁺ ions, thereby releasing His-tagged FAST-PETase from the matrix.

Protein elution was monitored by absorbance at **280 nm (A280)**, and fractions corresponding to the major protein peak were collected for subsequent quality assessment and functional characterization.

Importantly, the same general purification strategy was applied to WT FAST-PETase and the designed mutants. This reduced purification-related variability and allowed differences observed during subsequent activity measurements to be interpreted more confidently in relation to the introduced mutations.

Buffer Optimization and Batch Documentation

As the experimental workflow developed, the purification conditions were adjusted between different experimental batches.

The initial purification workflow employed phosphate/NaCl-based buffers, together with imidazole-containing wash and elution solutions. During later scale-up experiments, a Tris-HCl/NaCl/imidazole-based buffer system was used.

Rather than treating these conditions as interchangeable versions of a single protocol, we documented them as batch-specific purification conditions. For each protein-production batch, the buffer composition, pH, salt concentration, imidazole concentration, and relevant purification conditions were recorded together with the corresponding protein samples.

This distinction was important because buffer composition can influence protein solubility, nonspecific interaction with the affinity matrix, purification yield, and protein stability. Consequently, direct comparisons between FAST-PETase variants were performed using proteins prepared under matched conditions whenever possible.

Recording these changes also allowed optimization of the purification procedure itself to become part of our engineering process. Instead of silently replacing the initial protocol with the later one, we retained the experimental history so that changes in protein recovery or downstream behavior could be interpreted in the context of the conditions under which each batch was prepared.

Protein Quality Assessment and Preparation for Activity Testing

Following Ni^{2+} affinity purification, wild-type FAST-PETase and mutant proteins were prepared for quality assessment before being introduced into PET hydrolysis experiments.

Protein purity and apparent molecular weight were evaluated using SDS-PAGE, providing a direct check that recombinant FAST-PETase had been successfully expressed and enriched during purification. Protein concentration was subsequently determined using the corresponding quantification procedure so that enzyme concentrations could be normalized before functional comparison.

This quality-control stage was essential because apparent differences in PET degradation could otherwise arise from unequal amounts of enzyme rather than genuine differences in catalytic performance.

We therefore treated protein production and catalytic characterization as separate experimental checkpoints:

Successful plasmid construction does not necessarily imply successful protein expression, and successful protein expression does not necessarily imply improved enzymatic activity.

A designed mutation could influence protein expression, folding, solubility, stability, substrate recognition, or catalytic activity. By evaluating protein quality before the PET hydrolysis assay, we could distinguish these possibilities more carefully.

At this point, the complete workflow connected our computational predictions with experimentally testable enzyme preparations:

Mutation design → Molecular docking and MD analysis → Candidate selection → Mutant plasmid construction → Sequence verification → Recombinant expression → Ni²⁺ affinity purification → Protein quality control → PET hydrolysis assay

The purified proteins therefore provided the physical basis for determining whether structural and dynamic changes predicted *in silico* translated into measurable changes in PET hydrolysis.

Production and Purification of pET-32a/TrxA-Tagged FAST-TM Variants

In addition to the pET-22b(+)-based FAST-PETase expression workflow described above, selected variants were also produced using a pET-32a-derived construct containing a TrxA/6×His fusion region. Because this expression system required fusion-tag removal, its purification sequence differed from the direct His-tag purification workflow and was therefore documented separately.

Mutant constructs were generated by site-directed mutagenesis using the corresponding pET-32a template. PCR products were examined by agarose gel electrophoresis, and a DNA band consistent with the expected recombinant construct size was obtained. Candidate constructs were subsequently verified by sequencing before being carried forward into expression. In the recorded FAST-Y87Q-related construct workflow, the pET-32a vector and FAST-PETase coding region together produced an expected recombinant plasmid of approximately 6.7 kb, consistent with the observed electrophoretic band.

Following recombinant expression and cell disruption, soluble protein was subjected to primary Ni²⁺-affinity chromatography. SDS-PAGE analysis of the cell pellet, soluble supernatant, flow-through, wash, and elution fractions was used to follow enrichment of the recombinant fusion protein. A prominent band near approximately 45 kDa was observed in the soluble and enriched fractions, consistent with the approximate size expected for the FAST-TM protein together with the pET-32a-derived TrxA/His-containing fusion region. Fractions containing the target fusion protein were pooled for subsequent tag removal.

The pooled sample was then treated with TEV protease to remove the fusion region. SDS-PAGE analysis before and after cleavage showed a shift from the higher-molecular-weight fusion species toward a band near approximately 29 kDa, consistent with removal of the TrxA/His-containing region from the target protein. The cleaved preparation was subsequently subjected to a second nickel-affinity chromatography step to separate the untagged target protein from residual His-containing fusion components and other impurities.

For the recorded FAST-TM purification workflow, SDS-PAGE after the second affinity step showed a strong target-associated band near approximately 29 kDa in the relevant collected fraction, indicating further enrichment after cleavage and secondary purification. The purified protein was subsequently concentrated and prepared for downstream characterization.

A corresponding secondary nickel-affinity purification was also recorded for the FAST-S187Y-containing construct using stepwise imidazole concentrations ranging from 5 to 500 mM. SDS-PAGE was used to identify target-containing fractions, after which selected fractions were combined and concentrated by ultrafiltration for subsequent analysis.

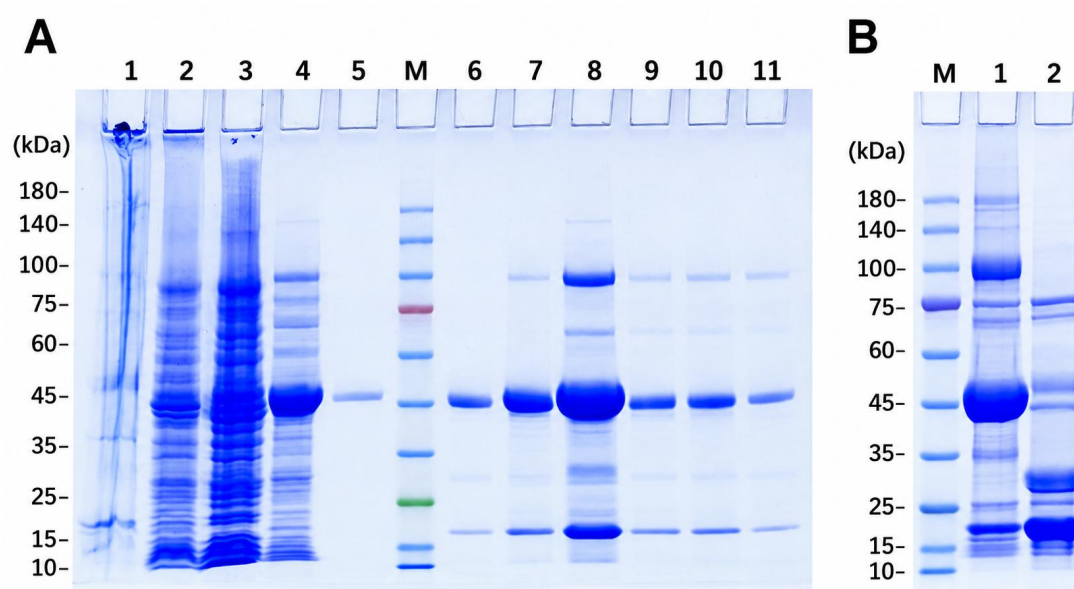


Figure: Primary nickel-affinity chromatography and TEV cleavage analysis of FAST-Y87Q

(A) SDS-PAGE analysis of protein expression and primary nickel-affinity purification. M, protein molecular-weight marker; lane 1, cell pellet/insoluble fraction; lane 2, soluble supernatant; lane 3, flow-through; lane 4, wash fraction; lane 5, 100% buffer B elution; lanes 6–11, representative fractions collected during purification. (B) SDS-PAGE analysis before and after TEV protease digestion. M, protein molecular-weight marker; lane 1, protein sample before TEV digestion; lane 2, protein sample after TEV digestion.

Secondary nickel-column affinity chromatography of FAST-S187Y

Secondary nickel-column affinity chromatography was performed using a gradient elution buffer consisting of 5 mM, 10 mM, 20 mM, 50 mM, and 500 mM imidazole. SDS-PAGE analysis revealed that the target protein in lanes 2–3 (the load flow-through) and the nickel column wash flow-through (the wash flow-through) was relatively pure, while the target protein in lanes 4–6 (the

wash flow-through) was less pure and contained a significant number of contaminants. Therefore, the load flow-through and nickel column wash flow-through were collected and concentrated by ultrafiltration.

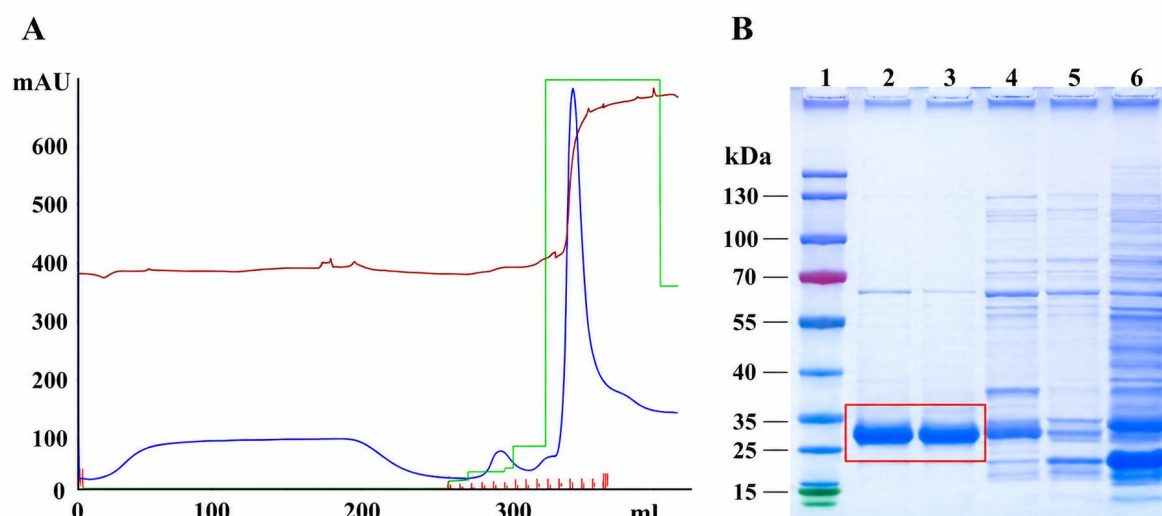


Figure: The 2nd Ni column affinity chromatography of FAST-S187Y. B: SDS-PAGE analysis. Lane 1: marker; 2: flow-through; 3: flush; 4-6: fractions of gradient elution; (Note: the red frame indicates the target band).

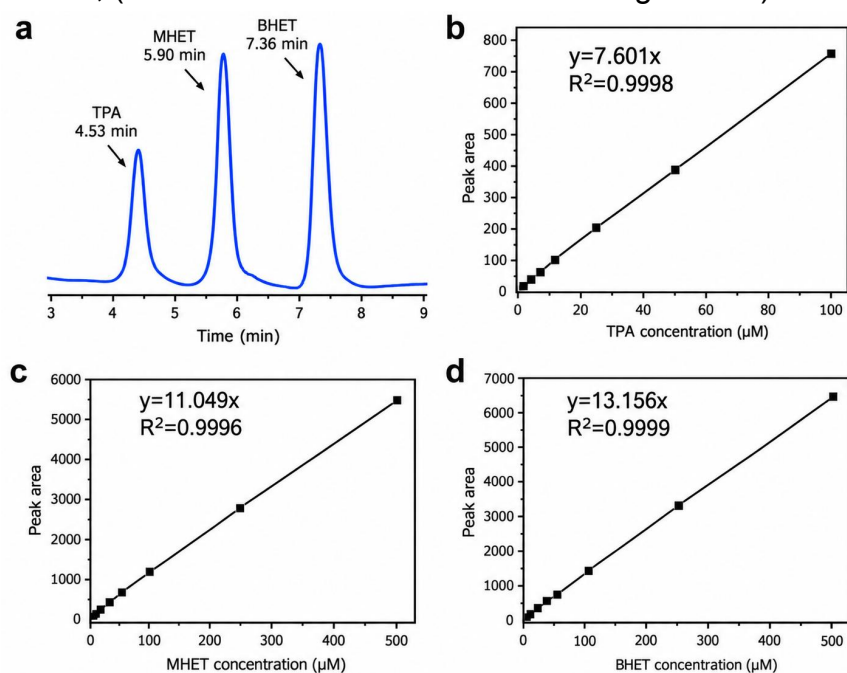


Figure: Typical HPLC charts and standard curves of TPA, MHET, and BHET. (a) Typical HPLC charts of 0.1 mM TPA, MHET, and BHET. The standard curves of (b) TPA, (c) MHET, and (d) BHET.

Week 15| 9.16~9.23

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Establishment and Validation of the PET Hydrolysis Assay

Experimental Validation of an Absorbance-Based PET Hydrolysis Assay

With purified PETase proteins available, the next challenge was to establish a practical method for comparing PET hydrolysis activity.

Conventional chromatographic methods such as UPLC provide direct quantification of soluble PET hydrolysis products, including MHET and TPA, but are relatively time-consuming when many enzyme variants or experimental conditions need to be compared.

We therefore evaluated whether bulk ultraviolet absorbance at 260 nm (A260) could serve as a simplified readout for PET hydrolysis.

The rationale was that soluble aromatic PET degradation products contribute to absorbance in this spectral region. As PET is hydrolyzed and soluble aromatic products accumulate in solution, A260 should therefore provide a convenient proxy for the overall extent of hydrolysis.

However, A260 does not distinguish individual products and cannot automatically be assumed to provide the same quantitative information as chromatographic analysis. We therefore first asked whether A260 could reliably reproduce relative differences in PET hydrolysis activity before applying it as a screening method.

A260 as a Rapid Screening Readout

To determine whether absorbance at 260 nm could provide a practical readout for rapid PETase screening, we compared four PETase enzymes representing different levels of PET-hydrolysis performance: wild-type PETase (WT), ThermoPETase (TP), FAST-PETase (FP), and PET hydrolysis assays were performed at 40 °C using enzyme concentrations of 150 nM and 300 nM. For each condition, hydrolysis was evaluated using two complementary analytical approaches. UPLC was used to quantify major soluble PET-hydrolysis products, particularly MHET and TPA, whereas A260 was measured as a simple proxy for the accumulation of soluble aromatic products. In this way, the absorbance-based method was evaluated against a product-specific chromatographic measurement rather than being assumed to represent PET conversion directly.

After 72 h of hydrolysis, A260 reproduced the same overall activity ranking observed by UPLC:

WT < TP < FP

This agreement indicated that A260 could distinguish PETase variants with substantially different hydrolysis activities under the tested conditions. Importantly, however, the agreement was stronger for relative ranking than for absolute product concentration. When A260 values were converted into MHET-equivalent concentrations, the absorbance-based method systematically gave higher apparent product concentrations than UPLC, with the extent of overestimation varying with enzyme identity and concentration. The discrepancy was approximately 16–52% across the tested conditions.

Despite this systematic difference in absolute concentration, the relative activity differences between enzymes were preserved remarkably well. At an enzyme concentration of 150 nM after 72 h, A260 gave relative activities of approximately 4.1-fold for TP, 6.5-fold for FP, compared with WT. The corresponding UPLC-derived values were approximately 4.0-fold, 6.6-fold, and 7.9-fold, respectively. Thus, the two analytical approaches reached essentially the same conclusion regarding the relative performance of the engineered PETases even though they did not report identical absolute concentrations.

We therefore used A260 as a rapid relative screening readout rather than an absolute product-quantification method. This distinction was important for the subsequent engineering workflow. A260 allowed multiple variants or conditions to be compared efficiently, whereas promising candidates could subsequently be examined using chromatographic methods capable of resolving individual PET-hydrolysis products.

The assay therefore established a two-level experimental strategy: rapid screening by A260 → product-specific confirmation by chromatography. This approach retained the throughput advantage of absorbance-based screening while reducing the risk of interpreting a nonspecific optical signal as direct quantitative evidence for formation of a particular PET-hydrolysis product.

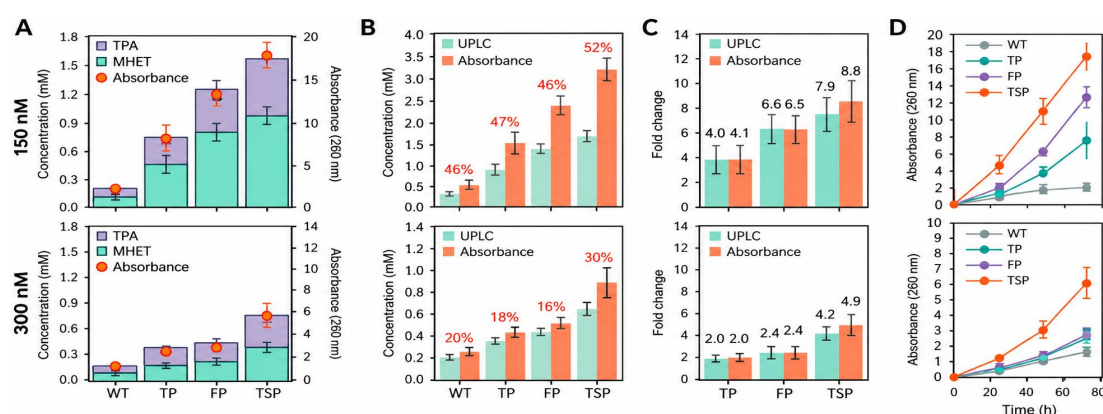


Figure. Validation of A260 as a rapid relative screening readout for PETase activity. Comparison of absorbance-based and UPLC-based measurements for WT

PETase, ThermoPETase (TP), FAST-PETase (FP), at 150 and 300 nM enzyme concentrations. A260 systematically overestimated absolute product concentration but reproduced the relative activity ranking observed by UPLC, supporting its use for first-pass mutant screening rather than absolute product quantification.

These experiments validate A260 as a rapid screening readout for PETase engineering. While absorbance measurements cannot replace UPLC for precise quantification of individual products such as TPA and MHET, they reliably capture relative activity differences, fold-changes, and time-dependent hydrolysis trends among PETase variants. This makes the absorbance-based assay particularly useful for screening large mutant libraries, where promising variants can first be identified by A260 and subsequently validated by more quantitative analytical methods such as UPLC. The comparison also identifies TSP as the most active benchmark variant under the tested conditions, providing an appropriate reference for evaluating newly engineered PETase mutants.

Chromatographic Calibration and Product-Specific Validation

Absorbance-based measurements provide a convenient measure of the accumulation of soluble aromatic material, but they do not distinguish between individual PET-hydrolysis products. We therefore used chromatographic analysis as an orthogonal method for product-specific validation.

Authentic standards of TPA, MHET, and BHET were analyzed to establish their characteristic chromatographic behavior and generate calibration relationships between analyte concentration and integrated peak area. These calibration curves provided the basis for converting chromatographic signals into product concentrations and allowed the major soluble products of PET hydrolysis to be evaluated independently. The corresponding chromatograms showed clearly resolved signals for TPA, MHET, and BHET, supporting their use as reference compounds for analysis of enzymatic reaction mixtures.

Chromatography therefore served a different role from A260 in our experimental workflow. Whereas A260 was used primarily to compare the overall relative performance of multiple variants rapidly, chromatographic analysis was used when information about product identity and composition was required.

The two measurements should consequently not be considered interchangeable. Instead, they provide complementary levels of experimental evidence:

A260 reports relative accumulation of UV-absorbing soluble products, whereas chromatography identifies and quantifies individual hydrolysis products.

This distinction allowed high-throughput screening and detailed product characterization to be incorporated into the same experimental workflow without requiring every initial variant to undergo full chromatographic analysis.

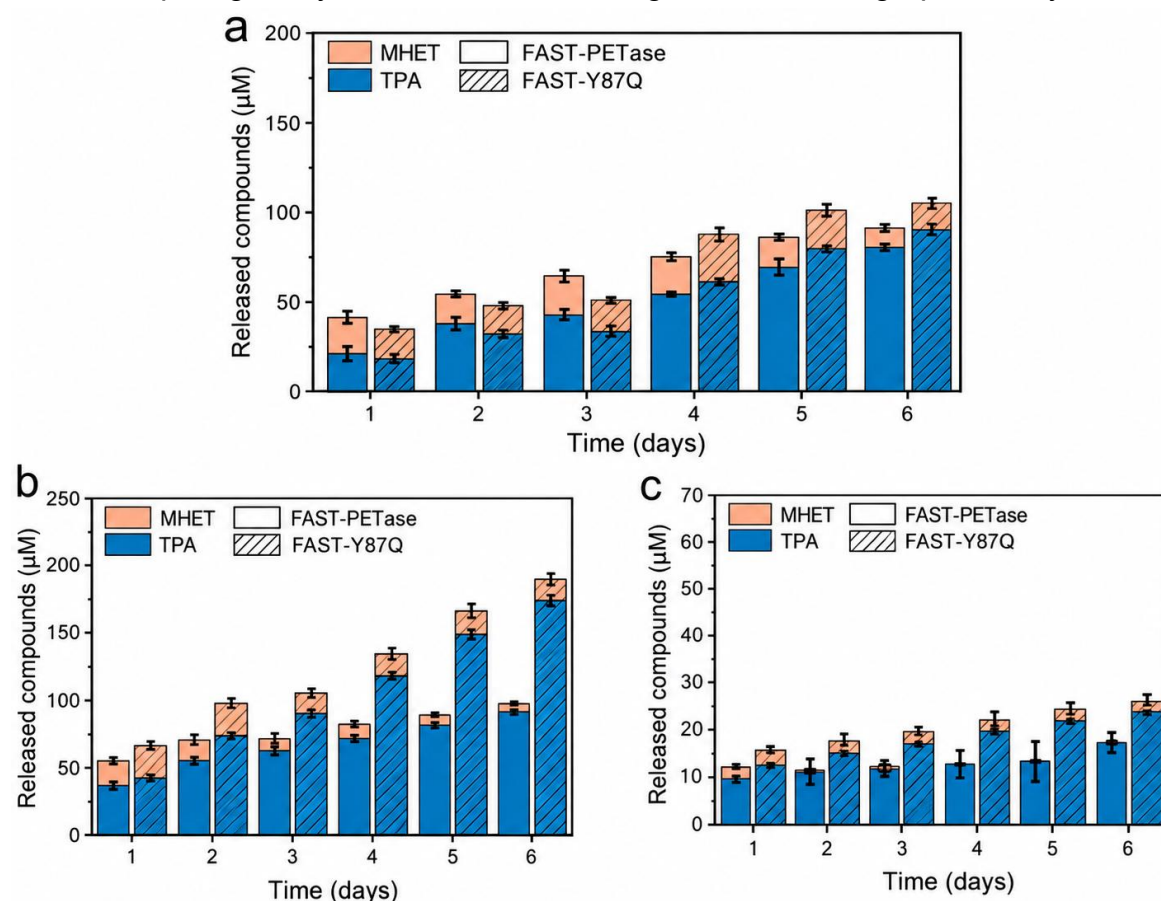


Figure. Time-dependent PET-hydrolysis product profiles of FAST-PETase and FAST-Y87Q.

TPA and MHET were quantified chromatographically during the hydrolysis time course. Both enzymes produced predominantly TPA with smaller amounts of MHET, and no major difference in overall product composition was observed between FAST-PETase and FAST-Y87Q under the tested conditions.

Week 16| 9.24~9.30

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Closing Perspective: From Mutation Discovery to Bottleneck-Guided Engineering

This Notebook records the development of an integrated workflow for PET-hydrolase engineering that connects computational design, molecular simulation, protein construction, functional testing, and mechanistic interpretation.

The project began with a simple engineering question: which mutations should be tested? Two complementary candidate-generation strategies were used to reduce the mutation space. Structure-aware graph learning provided one route for identifying potentially informative substitutions, whereas structure-based active-site design provided a second route focused more directly on the enzyme–substrate environment. In both cases, computational ranking was treated as a method for generating testable hypotheses rather than as a substitute for experimental validation.

As the project progressed, the central question became more mechanistic. Docking could suggest whether PET-derived substrates adopted plausible binding modes, but it could not establish catalytic competence. Molecular dynamics provided access to the conformational ensemble, while PRS analysis identified the subset of configurations compatible with nucleophilic attack. Quantum-chemical calculations then addressed a different question: once such a reaction-ready configuration had been reached, how energetically difficult was progression through the modeled chemical step? These levels of analysis were therefore complementary rather than interchangeable. As summarized earlier in the Notebook, MD determines which conformations are sampled, trajectory analysis identifies which may be reaction-ready, and DFT evaluates the energetic accessibility of chemical transformation from those states.

The experimental workflow provided the corresponding functional test. Designed constructs were sequence-verified, expressed, purified, and subjected to PET-hydrolysis assays under controlled conditions. The comparison between A260 and chromatographic measurements also highlighted an important experimental principle: a rapid screening signal can be highly useful without being an exact analytical measurement. A260 preserved relative activity differences among benchmark PETases, whereas chromatography provided product-specific confirmation. Together, these measurements allowed computationally proposed variants to be evaluated as physical enzymes rather than remaining purely in silico predictions.

The main conceptual lesson from this work is that improved PET hydrolysis cannot be assigned automatically to one universal descriptor. A mutation may influence substrate recognition, conformational sampling, reaction-ready geometry, local catalytic organization, transition-state energetics, protein stability, or several of these factors simultaneously.

Consequently, docking score, global RMSD, PRS population, calculated activation barrier, and experimental activity should not be expected to rank variants identically.

This observation motivates the bottleneck-guided view that underlies the project. Rather than asking only whether a variant is “better,” the more useful engineering question is:

Which step still limits productive catalysis in the starting scaffold, and which descriptor is most appropriate for testing that limitation?

When reaction-ready configurations are rarely sampled, engineering may benefit from mutations that improve substrate positioning and geometric preorganization. When such configurations are already accessible, further improvement may depend increasingly on their chemical organization, including proton-transfer readiness, oxyanion stabilization, electrostatic preorganization, or transition-state interactions.

The value of the Notebook is not only the individual mutations or calculations recorded within it, but the establishment of a traceable route from prediction to experiment and from phenotype to mechanism. Future rounds of engineering can build on this framework by testing additional variants, expanding the range of PET-hydrolase scaffolds, and refining the mechanistic descriptors used at each stage of the catalytic process.

Ultimately, the goal is not to optimize every measurable property simultaneously, but to use the appropriate experimental and computational evidence to identify—and then relieve—the bottleneck that matters most for the enzyme being engineered.

Appendix A — Experiment and Calculation Record Index

A.1 Record-Keeping and Traceability

This Notebook was prepared as a structured record of the experimental and computational work carried out during the project. Wherever possible, major experiments and calculations were linked to the corresponding experimental notes, computational files, analysis outputs, and figures used in the final project presentation.

The research records include wet-lab activities such as plasmid construction, sequence verification, recombinant protein expression, purification, PET-hydrolysis assays, absorbance measurements, chromatographic analysis, and protein-quality assessment. Computational records include mutation design, molecular docking, molecular dynamics simulations, trajectory analysis, pre-reaction-state analysis, QM-cluster calculations, and electronic-structure analysis.

Dates, researcher contributions, experimental conditions, computational parameters, and relevant output files are retained in the underlying project records where available. These records provide additional context for the

summarized workflows presented in this Notebook and allow important results to be traced back to their corresponding experimental or computational source.

For the final Wiki, selected records may additionally be linked to the figures and conclusions that they support in the iDEC Report.

Appendix B — Data and Code Availability

B.1 Data and Code Organization

Experimental and computational data generated during this project were organized alongside the corresponding Notebook sections and analysis workflows. These materials include PET-hydrolysis measurements, absorbance data, chromatographic calibration and product-analysis data, protein-expression and purification records, processed molecular-dynamics results, pre-reaction-state analyses, QM calculations, and supporting figure-source data.

Computational input files and analysis materials include molecular-docking configurations, Amber input files, CPPTRAJ analysis scripts, PRS-analysis procedures, Gaussian input and output files, and supporting wavefunction-analysis data.

Where appropriate, selected datasets, analysis scripts, and source files will be made available through the project Wiki or associated data/code repositories. Repository links, archived releases, or persistent identifiers can be added as these resources are finalized.

Experimental data

PET-hydrolysis measurements, A260 screening data, chromatographic calibration and product analysis, SDS-PAGE images, and protein-purification records.

Molecular simulation data

Prepared structures, molecular-dynamics inputs, processed trajectory analyses, PRS statistics, and representative structures.

Quantum-chemical data

QM-cluster structures, Gaussian calculations, transition-state and energy data, and supporting Multiwfn analyses.

Analysis resources

Relevant scripts, analysis definitions, plotting files, and figure-source data used during interpretation of the results.

B.2 Availability

Supporting data and computational resources are being organized for long-term project documentation. Where repository deposition is practical, links to the corresponding datasets or source-code archives will be provided through the Wiki.

Additional project records and biological materials may be available from the team upon reasonable request, subject to institutional requirements, data-storage limitations, software licensing, biosafety considerations, and material availability.

Appendix C — Data Accessibility Statement

Data Accessibility Statement

All raw and processed data supporting the conclusions of this project are available through the research-data repository listed above. These materials include experimental PET-hydrolysis measurements, chromatographic data, protein-expression and purification records, processed molecular-dynamics outputs, pre-reaction-state analyses, quantum-chemical results, and figure-source data.

Computational input files, analysis scripts, and source code used in molecular docking, molecular-dynamics analysis, PRS identification, quantum-chemical analysis, and figure preparation are available through the associated source-code repository and archived release.

Biological materials generated during this project, including the corresponding plasmid constructs and available expression strains, are available from the team upon reasonable request, subject to institutional requirements, biosafety regulations, material-transfer restrictions, and availability.

No personally identifiable research-participant data were generated in this project. Where repository deposition or sharing is subject to institutional, legal, licensing, or data-protection restrictions, these limitations are stated explicitly in the corresponding repository record.